Measure-to-measure interpolation using Transformers

A machine-checked blueprint: the original statements and proofs, our Lean formalization, and what it strengthens

Alexandre Quemy

Table of contents

About this blueprint

This is a standalone, A-to-Z blueprint of the controllability results of

> M. Geshkovski, P. Rigollet, Y. Ruiz-Balet, *Measure-to-measure interpolation using Transformers*, arXiv:2411.04551v3,

together with their machine-checked Lean 4 formalization. It is meant to be read on its own: for each node it reproduces the **original statement (and proof, where short) verbatim**, then gives **our formalization** (a faithful statement in Lean, a prose proof, the real source listing, and an honest status badge), and, where relevant, **what our work strengthens** (a corrected sign error, a discharged rigor gap, or exactly what is axiomatised).

Authorship and attribution. The mathematics, the theorems, and the proofs are the work of the original authors. The Lean formalization, the numerical experiments, the prose informalizations, and this blueprint are the work of Alexandre Quemy. Original statements and proofs are reproduced verbatim, in clearly marked grey boxes with a page-cited attribution, for scholarly comparison (fair use); all such material remains the authors'.

How to read the boxes.

From the original paper. Grey boxes reproduce the authors' statements and proofs verbatim, with a page-cited attribution.

Our formalization. Blue boxes hold our Lean statement, a prose proof, the real source listing, and a status badge: **machine-checked** (clean Lean kernel proof), **axiomatised** (rests on a labeled axiom standing in for missing Mathlib theory), **open** (stated in Lean as a `sorry`), or **informal** (reviewed on paper).

What we strengthen. Green boxes record where the formalization corrects or completes the paper: a sign error (F1), a rigor gap discharged by a kernel-checked leaf (F2), or what had to be axiomatised.

The status badge of every machine-checked node is verified against the Lean kernel by `#print axioms` (`bin/axiom-report`): a node is badged machine-checked only if its axiom set is clean (`propext`, `Classical.choice`, `Quot.sound` only).

1 Foundations

The paper views a Transformer as the flow map of a continuity equation on the unit sphere $\mathbb{S}^{d-1}$. We reproduce the setting verbatim, then give the kernel-checked algebraic foundations our leaves rest on.

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §1, pp. 2-3).
Viewing each token as a particle, given an initial sequence $(x_1(0), \dots, x_n(0)) \in (\mathbb{S}^{d-1})^n$, one considers

$$\dot{x}_i(t) = \mathsf{v}[\mu(t)](t, x_i(t)), \qquad t \in [0, T], \tag{1.1}$$

where $\mu(t) = \frac{1}{n}\sum_{j=1}^n \delta_{x_j(t)}$ is the empirical measure. The vector field is

$$\mathsf{v}[\mu](t,x) = \mathbf{P}_x^\perp(\mathbf{V}(t)\,\mathcal{A}_{\mathbf{B}}[\mu](t,x) + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+), \tag{1.2}$$

where the self-attention is

$$\mathcal{A}_{\mathbf{B}}[\mu](t,x) := \int e^{\langle \mathbf{B}(t)x, x'\rangle} x'\,\mu(\mathrm{d}x') \Big/ \int e^{\langle \mathbf{B}(t)x, \zeta\rangle} \mu(\mathrm{d}\zeta).$$

The vector field is "ultimately projected onto $\mathsf{T}_x\mathbb{S}^{d-1}$ by virtue of the orthogonal projector $\mathbf{P}_x^\perp := I_d - xx^\top$, referred to as *layer normalization*." Since (1.1) depends only on $\mu(t)$, one passes to the **continuity equation**

$$\partial_t\mu(t) + \mathrm{div}(\mu(t)\,\mathsf{v}[\mu(t)]) = 0 \ \text{ on } [0,T] \times \mathbb{S}^{d-1}, \qquad \mu(0) = \mu_0. \tag{1.3}$$

For any control $\theta = (\mathbf{V}, \mathbf{B}, \mathbf{W}, \mathbf{U}, b)$ the Cauchy problem (1.3) is well-posed and yields a continuous, invertible flow map $\Phi_\theta^t : \mathcal{P}(\mathbb{S}^{d-1}) \to \mathcal{P}(\mathbb{S}^{d-1})$ with $\Phi_\theta^t(\mu_0) = \mu(t)$.

1.1 Tangential projector (layer normalization)

Our formalization. machine-checked
We define the layer-normalization projector $\mathbf{P}_x^\perp v = v - \langle x, v\rangle x$ and prove the algebraic identities every leaf uses (ledger node **L1**): it is linear, symmetric, idempotent on the sphere, annihilates x, and satisfies the quadratic identity

$$\langle \mathbf{P}_x^\perp v, v\rangle = \|v\|^2 - \langle x, v\rangle^2 \qquad (\|x\| = 1),$$

which is exactly the scalar driving the gate ODE (B.5) and the barycenter ODE (B.9). The proof is a direct computation: expanding $\langle v - \langle x, v\rangle x,\ v\rangle$ and using $\langle x, x\rangle = 1$ cancels the cross term to $\|v\|^2 - \langle x, v\rangle^2$.

```
/-- The tangential projector `P_x^⊥ v = v - ⟪x, v⟫ x`. -/
noncomputable def tangentialProjector (x v : Eucl d) : Eucl d := v - (⟪x, v⟫) • x

/-- The key quadratic identity (L1): for a unit vector `x`,
`⟪P_x^⊥ v, v⟫ = ‖v‖² - ⟪x, v⟫²`. -/
theorem projector_inner_sub_sq {x : Eucl d} (hx : x ∈ sphere d) (v : Eucl d) :
    ⟪tangentialProjector x v, v⟫ = ‖v‖ ^ 2 - ⟪x, v⟫ ^ 2 := by
  have h : ⟪x, x⟫ = 1 := inner_self_eq_one_of_mem_sphere hx
  simp only [tangentialProjector, inner_sub_left, inner_smul_left, RCLike.conj_to_real,
    real_inner_self_eq_norm_sq]
  ring
```

Foundations/Projector.lean

1.2 Geodesic distance

Our formalization. machine-checked

The geodesic distance on the sphere is $d_g(x,y) = \arccos\langle x,y\rangle$. We prove the inversion $\cos d_g(x,y) = \langle x,y\rangle$ for unit vectors and that $d_g(x,y) \in [0,\pi]$. These let the gate sign (B.4) be read either as the inner-product inequality $\langle z,x\rangle < \cos R$ or as the geodesic statement $d_g(z,x) > R$ (via strict antitonicity of cos on $[0,\pi]$).

```
/-- Geodesic distance on the unit sphere: `d_g(x, y) = arccos⟪x, y⟫`. -/
noncomputable def geodesicDist (x y : Eucl d) : ℝ := Real.arccos (⟪x, y⟫)

/-- `cos (d_g x y) = ⟪x, y⟫`: the cosine of the geodesic distance is the inner product.
 ↪ -/
theorem cos_geodesicDist {x y : Eucl d} (hx : x ∈ sphere d) (hy : y ∈ sphere d) :
    Real.cos (geodesicDist x y) = ⟪x, y⟫ :=
  Real.cos_arccos (neg_one_le_inner hx hy) (inner_le_one hx hy)
```

Foundations/GeodesicDistance.lean

1.3 Geodesic convexity and the separation transfer (M5)

Our formalization. machine-checked

The Section 3.3 geometry the paper uses without proof, fully kernel-checked: strict spherical caps at nonnegative level are geodesically convex (`geodesicConvex_inner_cap`), polarization bridges identify Euclidean `r`-balls with caps at level $1 - r^2/2$, and the **separation transfer** – measures shrunk into `r`-balls around `2r`-separated unit directions have pairwise disjoint geodesic hulls and pairwise non-colinear barycenters (`geodesicHull_disjoint_of_separated_balls`, `barycenter_not_sameRay_of_separated_balls`, with the strict barycenter-level bound `inner_barycenter_gt` via an a.e. vanishing argument). With M5 done, the disentanglement axiom owes only mean-field dynamics, no geometry.

1.4 The mean-field attention layer (M3b)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, eq. (1.2), pp. 2-3).

$$\mathsf{v}[\mu](t,x) = \mathbf{P}_x^{\perp}(\mathbf{V}(t)\,\mathcal{A}_{\mathbf{B}}[\mu](t,x) + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+), \qquad \mathcal{A}_{\mathbf{B}}[\mu](t,x) := \frac{\int e^{\langle \mathbf{B}(t)x,\, x'\rangle}\, x'\, \mu(\mathrm{d}x')}{\int e^{\langle \mathbf{B}(t)x,\, \zeta\rangle}\, \mu(\mathrm{d}\zeta)}.$$

The self-attention average $\mathcal{A}_{\mathbf{B}}[\mu]$ integrates against the *evolving* measure: the velocity field depends on $\mu(t)$ itself, which is precisely what eq. (1.7) shows a linear continuity equation cannot replicate.

Our formalization. axiomatised
The field (1.2) is a concrete **definition**: `AttnParams` packages the five parameters $(\mathbf{V}, \mathbf{B}, \mathbf{W}, \mathbf{U}, b)$ of one constant piece, `attnAvg` is the softmax average $\mathcal{A}_{\mathbf{B}}$, and `AttnParams.field` assembles the projected velocity. A mean-field flow is the predicate `IsMeanFieldFlow` (initial condition, measurability, Lipschitz regularity, sphere bijectivity, and the characteristic ODE against the *pushforward-evolved* measure), and schedules compose by list concatenation (`AttnSchedule`, `attnMeasureFlow`, `durationSum`, `switches` counting constant pieces). Exactly one axiom pair remains – existence and uniqueness of the mean-field flow for sphere-supported probability data (`exists_meanFieldFlow`, `meanFieldFlow_unique`): McKean-Vlasov well-posedness on the sphere, the honest remaining M3 gap. Every measure-dependent statement of the paper is stated over this layer; the linear `Block` flow hosts the paper's $\mathbf{V} \equiv 0$ sections. The two layers are bridged, kernel-checked on the predicate side: `isMeanFieldFlow_blockFlow` shows a linear `Block` matching a $\mathbf{V} = 0$ attention field on the sphere satisfies the mean-field predicate, and `attnStep_eq_map_blockFlow` (the first consumer of `meanFieldFlow_unique`) pins the attention step to the linear pushforward – so the discharged Appendix-B machinery acts on this layer too.

```
/-- One constant-in-time Transformer block (paper eq. (1.2)). -/
structure AttnParams (d : ℕ) where
  V B W U : Eucl d →L[ℝ] Eucl d
  b : Eucl d
  duration : ℝ
  duration_nonneg : 0 ≤ duration

/-- The measure-valued softmax average `A_B[μ](x)`. -/
noncomputable def attnAvg (B : Eucl d →L[ℝ] Eucl d) (μ : Measure (Eucl d)) (x : Eucl d)
 ↪  : Eucl d :=
  (∫ z, Real.exp ⟪B x, z⟫ ∂μ)⁻¹ • ∫ z, Real.exp ⟪B x, z⟫ • z ∂μ

/-- Mean-field well-posedness (McKean-Vlasov on the sphere): the single remaining
infrastructure axiom. -/
axiom exists_meanFieldFlow (p : AttnParams d) (μ₀ : Measure (Eucl d))
    [IsProbabilityMeasure μ₀] (hs : μ₀ (sphere d)ᶜ = 0) :
    ∃ Φ, IsMeanFieldFlow p μ₀ Φ
```

Foundations/Attention.lean

2 Self-contained leaves

These are the self-contained computational cores the paper's proofs rest on, each isolated from its measure-theoretic and flow-map scaffolding and given a clean Lean kernel proof. Two of them (L7, L8) reduce instead to a single labeled Wasserstein axiom and are flagged as such; the gate leaf (L2) and the non-colinearity leaves (L11, L11′) additionally repair or discharge a gap in the original text.

2.1 Gate algebra and gate ODE

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2 and eq. (B.4)-(B.5), pp. 31-32).
Lemma B.2 transports mass between two overlapping balls $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$ ($\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$): for any $\varepsilon > 0$, $T > 0$ there exist $W, U \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ so that the solution μ satisfies $\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1-\varepsilon)\mu_0(\mathcal{B}_0)$. In the proof:
"Let $z \in \mathbb{S}^{d-1}$ denote the center and $R > 0$ the radius of $\mathcal{B}_0$. Take an arbitrary $\omega \in \mathrm{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$. We consider $U = -\mathbf{1}z^\top$ and $b = \cos(R)\mathbf{1}$, as well as any $W \in \mathcal{M}_{d\times d}(\mathbb{R})$ such that $W\mathbf{1} = \omega$. Then $W(Ux+b)_+ = (-\cos d_g(z,x) + \cos(R))_+\,\omega$, and note that"

$$(-\cos d_g(z,x) + \cos(R))_+ > 0 \quad \iff \quad x \in \mathcal{B}_0. \tag{B.4}$$

"Now observe that"

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle x(t), \omega\rangle = (-\cos d_g(z, x(t)) + \cos(R))_+(1 - \langle x(t), \omega\rangle^2), \tag{B.5}$$

"which is positive whenever $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$."

Our formalization. machine-checked
The gated drift is $W(Ux+b)_+ = g \cdot \omega$ with scalar gate $g = (\cos R - \langle z, x\rangle)_+$. Projecting onto the tangent space and pairing with the unit vector ω, the projector identity L1 collapses the tangential part to a scalar: $\langle \mathbf{P}^\perp_x(g\,\omega), \omega\rangle = g\,\langle \mathbf{P}^\perp_x \omega, \omega\rangle = g(\|\omega\|^2 - \langle x, \omega\rangle^2) = g(1 - \langle x, \omega\rangle^2)$ since $\|\omega\| = 1$ — this is `gate_inner_identity`. The ODE (B.5) is then the chain rule: $\frac{\mathrm{d}}{\mathrm{d}t}\langle x(t), \omega\rangle = \langle x'(t), \omega\rangle$ (ω constant), and substituting the projected drift $x'(t) = \mathbf{P}^\perp_{x(t)}(g\,\omega)$ reproduces the logistic factor. For the sign (B.4), $0 < \cos R - \langle z, x\rangle \iff \langle z, x\rangle < \cos R$, and rewriting $\langle z, x\rangle = \cos d_g(z,x)$ with strict antitonicity of cos on $[0,\pi]$ gives the geodesic form. The key step is the quadratic projector identity, which turns the ambient projection into the scalar $1 - \langle x, \omega\rangle^2$.

```
/-- L2 (algebra of B.5): for unit `x` and unit `ω`, projecting the scaled drift `g•ω`
↪ and pairing
with `ω` gives the logistic factor `g(1 - ⟪x,ω⟫²)`. -/
theorem gate_inner_identity {x ω : Eucl d} (hx : x ∈ sphere d) (hω : ω ∈ sphere d) (g :
↪ ℝ) :
    ⟪tangentialProjector x (g • ω), ω⟫ = g * (1 - ⟪x, ω⟫ ^ 2) := by
  rw [tangentialProjector_smul, real_inner_smul_left, projector_inner_sub_sq hx ω,
    norm_eq_one_of_mem_sphere hω, one_pow]

/-- L2 (the gate ODE B.5): along a path `x(t)` on the sphere whose velocity is the
↪ projected gated
drift `x'(t) = P_{x(t)}^⊥ (g • ω)`, the inner product `⟪x(t), ω⟫` evolves as
`g(1 - ⟪x(t), ω⟫²)`. -/
theorem gate_hasDerivAt_inner {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hω : ω ∈ sphere d) (g : ℝ)
    (hode : x' = tangentialProjector (x t) (g • ω)) :
    HasDerivAt (fun s ⇒ (⟪x s, ω⟫ : ℝ)) (g * (1 - ⟪x t, ω⟫ ^ 2)) t := by
  have hconst : HasDerivAt (fun _ : ℝ ⇒ ω) (0 : Eucl d) t := hasDerivAt_const t ω
  have h := hx.inner ℝ hconst
  rw [hode, gate_inner_identity hxs hω g] at h
  simpa using h

/-- L2 (gate sign, B.4, geodesic form): the gate is active exactly outside the closed
↪ cap of radius
`R` around `z`, i.e. on `{x : d_g(z, x) > R}`. -/
theorem gate_pos_iff_dist {z x : Eucl d} (hz : z ∈ sphere d) (hx : x ∈ sphere d)
    {R : ℝ} (hR : R ∈ Set.Icc (0 : ℝ) Real.pi) :
    (⟪z, x⟫ : ℝ) < Real.cos R ↔ R < geodesicDist z x := by
  rw [← cos_geodesicDist hz hx]
  exact Real.strictAntiOn_cos.lt_iff_gt (geodesicDist_mem_Icc z x) hR
```

Leaves/GateODE.lean

Strengthening (finding F1). The printed (B.4) asserts the gate is active iff $x \in \mathcal{B}_0 = B(z, R)$. Our kernel-checked `gate_pos_iff_dist` proves instead $(\cos R - \langle z, x\rangle)_+ > 0 \iff d_g(z, x) > R$ — the gate is active on the **complement** of the cap. The opposite sign is the consistent one: (B.5) needs the gate active *inside* $\mathcal{B}_0 \setminus \{\omega\}$ to push interior mass toward ω, whereas the printed parameters $U = -\mathbf{1}z^\top$, $b = \cos(R)\mathbf{1}$ leave the interior mass on a zero gate, so the lemma as written transports nothing; the fix is $U = +\mathbf{1}z^\top$, $b = -\cos(R)\mathbf{1}$. This is corroborated independently by experiment E1 (the gated-transport mass fraction rises from ≈ 0.27 to 1.0 once the seed is placed on $\{d_g(z, x) > R\}$) and by the paper's own §4 Step 3, which states the identical construction with the correct side. See `ERRATA.md`.

2.2 Barycenter ODE

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, eq. (B.9), p. 33).
In Step 1 of the proof of Lemma 3.3, the solution to

$$\begin{cases} \partial_t \mu(t) + \operatorname{div}\big(\mathbf{P}_x^\perp \langle \alpha_1, \mathbb{E}_{\mu(t)}[x]\rangle\, \alpha_1\, \mu(t)\big) = 0 & \text{on } \mathbb{R}_{\geq 0} \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1} \end{cases} \tag{B.9}$$

for $\mu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ satisfies

$$\frac{\mathrm{d}}{\mathrm{d}t}\langle \mathbb{E}_{\mu(t)}[x], \alpha_1\rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1\rangle \Big(1 - \int \langle x', \alpha_1\rangle^2 \mu(t, \mathrm{d}x')\Big).$$

Our formalization. machine-checked
The disentangling field is $\mathbf{P}_x^\perp(c\,\alpha)$ with $c = \langle \mathbb{E}_\mu[x], \alpha\rangle$, so the per-particle characteristic $\langle x(t), \alpha\rangle$ obeys exactly the gate identity of L2 with c in place of the gate scalar: $\frac{\mathrm{d}}{\mathrm{d}t}\langle x(t), \alpha\rangle = c\,(1 - \langle x(t), \alpha\rangle^2)$. The Lean derivative is therefore literally `gate_hasDerivAt_inner` instantiated at $g = c$. The sign lemma `barycenter_deriv_pos` records that when $c > 0$ and the component lies strictly inside $(-1, 1)$ the right-hand side is positive, the dissipation that drives $\langle x, \alpha\rangle \to 1$, i.e. $x \to \alpha$. The key point is that the paper shows c is sign-preserved along the flow, so the logistic ODE is monotone.

```
/-- L6 (derivative, eq. B.9): along `x'(t) = P_{x(t)}^⊥ (c • α)` with `α` a unit vector,
↪ the
component `⟪x(t), α⟫` evolves as `c(1 - ⟪x(t), α⟫²)`. -/
theorem barycenter_hasDerivAt_inner {x : ℝ → Eucl d} {α : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (hxs : x t ∈ sphere d) (hα : α ∈ sphere d) (c : ℝ)
    (hode : x' = tangentialProjector (x t) (c • α)) :
    HasDerivAt (fun s => (⟪x s, α⟫ : ℝ)) (c * (1 - ⟪x t, α⟫ ^ 2)) t :=
  gate_hasDerivAt_inner hx hxs hα c hode

/-- L6 (sign / strict increase): when the barycenter coefficient `c` is positive and the
↪ component
is strictly inside `(-1, 1)`, the component strictly increases. This is the dissipation
↪ that pushes
`⟪x, α⟫` toward `1`. -/
theorem barycenter_deriv_pos {c y : ℝ} (hc : 0 < c) (hy : y ∈ Set.Ioo (-1 : ℝ) 1) :
    0 < c * (1 - y ^ 2) := by
  obtain ⟨h1, h2⟩ := hy
  have hy2 : y ^ 2 < 1 := by nlinarith
  have : 0 < 1 - y ^ 2 := by linarith
  positivity
```

Leaves/BarycenterODE.lean

2.3 Geodesic-distance gradient

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, eq. after (4.4), p. 20). "Observe that the trajectories of the ODE"

$$\dot{x}(t) = (\langle \gamma, x(t)\rangle - \epsilon/2)_+ \, \mathbf{P}^{\perp}_{x(t)}\omega_+ \qquad \text{for } t \geq 0, \tag{4.4}$$

"follow the Riemannian gradient flow of the distance between ω_+ and x in $\mathcal{S}_+$. Indeed,"

$$\nabla_1 d_g(x, \omega_+) = -\frac{\mathbf{P}^{\perp}_{x}(\omega_+)}{\sqrt{1 - \langle x, \omega_+\rangle^2}}.$$

Our formalization. machine-checked

Two facts pin down the gradient. First, the chain rule for $\arccos \circ \langle \cdot, \omega\rangle$: since $(\arccos)'(u) = -1/\sqrt{1-u^2}$ (Mathlib's `Real.hasDerivAt_arccos`), $\frac{\mathrm{d}}{\mathrm{d}t} d_g(x(t), \omega) = -(1 - \langle x, \omega\rangle^2)^{-1/2} \langle x', \omega\rangle$. Second, for a tangent velocity ($\langle x', x\rangle = 0$) the ambient ω may be replaced by its tangential projection: $\langle x', \mathbf{P}^{\perp}_x \omega\rangle = \langle x', \omega\rangle - \langle x, \omega\rangle\langle x', x\rangle = \langle x', \omega\rangle$. Hence the derivative is $\langle x', -(1 - \langle x, \omega\rangle^2)^{-1/2}\, \mathbf{P}^{\perp}_x \omega\rangle$, exhibiting $-\mathbf{P}^{\perp}_x \omega / \sqrt{1 - \langle x, \omega\rangle^2}$ as the Riemannian (tangential) gradient. The key step is that tangency annihilates the cross term, letting the Euclidean inner product against ω be read as the inner product against the gradient direction.

```
/-- L4 (derivative of geodesic distance along a path): the chain rule for `arccos∘⟪·,
 ↪  ω⟫`. -/
theorem geodesicDist_hasDerivAt {x : ℝ → Eucl d} {ω : Eucl d} {t : ℝ} {x' : Eucl d}
    (hx : HasDerivAt x x' t) (h1 : (⟪x t, ω⟫ : ℝ) ≠ -1) (h2 : (⟪x t, ω⟫ : ℝ) ≠ 1) :
    HasDerivAt (fun s ⇒ geodesicDist (x s) ω)
      (-(1 / Real.sqrt (1 - (⟪x t, ω⟫ : ℝ) ^ 2)) * ⟪x', ω⟫) t := by
  have hinner : HasDerivAt (fun s ⇒ (⟪x s, ω⟫ : ℝ)) (⟪x', ω⟫) t := by
    have h := hx.inner ℝ (hasDerivAt_const t ω)
    simpa using h
  have h := (Real.hasDerivAt_arccos h1 h2).comp t hinner
  simpa only [geodesicDist, Function.comp_def] using h

/-- L4 (gradient direction): on the sphere, for a tangent velocity the inner product
 ↪  with `ω` equals
the inner product with the tangential projection `P_x^⊥ ω`. Hence the geodesic-distance
 ↪  derivative
is `⟪x', -(1/√(1-⟪x,ω⟫²)) • P_x^⊥ ω⟫`, exhibiting `-P_x^⊥ ω / √(1-⟪x,ω⟫²)` as the
 ↪  Riemannian
gradient. -/
theorem inner_eq_inner_tangentialProjector {x ω x' : Eucl d}
    (htangent : (⟪x', x⟫ : ℝ) = 0) :
    (⟪x', ω⟫ : ℝ) = ⟪x', tangentialProjector x ω⟫ := by
  rw [tangentialProjector, inner_sub_right, real_inner_smul_right, htangent]
  ring
```

Leaves/GeodesicGradient.lean

2.4 Separating hyperplane

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 4.2 Step 1, p. 20). With ω_+ on the minimizing geodesic between ω and γ at distance $\pi/8$ from ω, one has $d_g(\omega_+, x_0^M) \geq d_g(\omega, x_0^M) - d_g(\omega, \omega_+) \geq \frac{3\pi}{8}$.
"As a consequence, the hyperplane $\{x \in \mathbb{S}^{d-1} : \langle \omega, x\rangle = \cos(\pi/8 + \tau)\}$ is a separating hyperplane for the ball $B(\omega, \pi/8 + \tau)$ and the points x_0^M and y^M for every $\tau \in (0, 3\pi/8)$; namely"

$$\langle \omega, x_0^M\rangle - \cos\Big(\frac{\pi}{8} + \tau\Big) = \cos d_g(\omega, x_0^M) - \cos\Big(\frac{\pi}{8} + \tau\Big) < 0,$$

"where the inequality is by virtue of (4.3)" (the choice $d_g(\omega, x_0^M) \geq \pi/2$).

Our formalization. machine-checked
The whole content is the strict antitonicity of cosine on $[0, \pi]$. From $\tau < 3\pi/8$ we get the cap angle $\pi/8 + \tau < \pi/2 \leq d_g(\omega, x)$, and both angles lie in $[0, \pi]$, so Mathlib's `Real.strictAntiOn_cos` gives $\cos d_g(\omega, x) < \cos(\pi/8 + \tau)$. Rewriting $\cos d_g(\omega, x) = \langle \omega, x\rangle$ lands the separating inequality $\langle \omega, x\rangle < \cos(\pi/8 + \tau)$. The key step is that monotonicity of cosine converts the angle gap $\pi/8 + \tau < d_g(\omega, x)$ into the inner-product gap that places x strictly on the far side of the hyperplane.

```
/-- L3 (separating side): if `x` is far from the anchor `ω` (`d_g(ω, x) ≥ π/2`) and
`τ ∈ (0, 3π/8)`, then `⟪ω, x⟫ < cos(π/8 + τ)`, so `x` lies strictly on the far side of
↪ the
separating hyperplane. -/
theorem separating_hyperplane {x ω : Eucl d} (hx : x ∈ sphere d) (hω : ω ∈ sphere d)
    {τ : ℝ} (hτ : τ ∈ Set.Ioo 0 (3 * Real.pi / 8))
    (hfar : Real.pi / 2 ≤ geodesicDist ω x) :
    ⟪ω, x⟫ < Real.cos (Real.pi / 8 + τ) := by
  obtain ⟨hτ0, hτ8⟩ := hτ
  have hpi := Real.pi_pos
  -- both angles lie in [0, π]
  have ha : Real.pi / 8 + τ ∈ Set.Icc (0 : ℝ) Real.pi := by
    constructor <;> nlinarith [Real.pi_pos]
  have hb : geodesicDist ω x ∈ Set.Icc (0 : ℝ) Real.pi := geodesicDist_mem_Icc ω x
  -- the far angle exceeds the cap angle
  have hab : Real.pi / 8 + τ < geodesicDist ω x := by nlinarith [hfar]
  -- cosine is strictly decreasing on [0, π]
  have hcos : Real.cos (geodesicDist ω x) < Real.cos (Real.pi / 8 + τ) :=
    Real.strictAntiOn_cos ha hb hab
  rwa [cos_geodesicDist hω hx] at hcos
```

`Leaves/SeparatingHyperplane.lean`

2.5 Lyapunov sign

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Example 6.1, pp. 29-30). On the circle ($d = 2$), for a particle following a characteristic of $\mu^i(t)$ one has $\dot{x}(t) = \mathbf{P}^{\perp}_{x(t)}(m_i(t)) = \alpha_i(t)\mathbf{P}^{\perp}_{x(t)}(u_i)$ and, in the angle variable $\vartheta := \theta - \theta_i$,

$$\dot{\vartheta}(t) = -\alpha_i(t) \sin \vartheta(t).$$

"The Lyapunov function $\mathsf{E}(\vartheta) := 1 - \cos \vartheta$ satisfies"

$$\dot{\mathsf{E}} = \sin \vartheta \, \dot{\vartheta} = -\alpha_i(t) \sin^2 \vartheta \leq 0,$$

"with equality only at $\vartheta = 0$ (since $\alpha_i(t) > 0$ ...). Thus $\vartheta(t) \to 0$ and each trajectory converges to u_i."

Our formalization. machine-checked
Both halves are kernel-checked. The derivative is the chain rule: $\frac{\mathrm{d}}{\mathrm{d}t}(1 - \cos \theta(s)) = \sin \theta(t) \, \theta'(t)$, and substituting the ODE $\theta' = -\alpha \sin \theta$ gives $-\alpha \sin^2 \theta$. The sign lemma `lyapunov_deriv_nonpos` then notes $-\alpha \sin^2 \theta \leq 0$ for $\alpha \geq 0$ because $\alpha \sin^2 \theta \geq 0$. So $\mathsf{E} = 1 - \cos \theta$ is nonincreasing along the angle flow, forcing $\theta \to 0$. The key step is that the $\sin \theta$ produced by the ODE cancels with the $\sin \theta$ from differentiating $1 - \cos \theta$, leaving a perfect square of definite sign.

```
/-- L5 (derivative of the Lyapunov function along the flow): if `θ` solves `θ' = -α sin
 ↪ θ` at `t`,
then `E(s) = 1 - cos θ(s)` has derivative `-α sin²(θ t)` at `t`. -/
theorem lyapunov_hasDerivAt (α : ℝ) (θ : ℝ → ℝ) (t θ' : ℝ)
    (hθ : HasDerivAt θ θ' t) (hode : θ' = -α * Real.sin (θ t)) :
    HasDerivAt (fun s => 1 - Real.cos (θ s)) (-α * Real.sin (θ t) ^ 2) t := by
  have hcos : HasDerivAt (fun s => Real.cos (θ s)) (-Real.sin (θ t) * θ') t :=
    (Real.hasDerivAt_cos (θ t)).comp t hθ
  have h : HasDerivAt (fun s => 1 - Real.cos (θ s)) (0 - (-Real.sin (θ t) * θ')) t :=
    (hasDerivAt_const t 1).sub hcos
  convert h using 1
  rw [hode]; ring

/-- L5 (sign): the Lyapunov derivative is nonpositive when `α ≥ 0`. -/
theorem lyapunov_deriv_nonpos {α : ℝ} (hα : 0 ≤ α) (s : ℝ) :
    -α * Real.sin s ^ 2 ≤ 0 := by
  have : 0 ≤ α * Real.sin s ^ 2 := mul_nonneg hα (sq_nonneg _)
  linarith
```

Leaves/Lyapunov.lean

2.6 Ball-chain retention

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1 and its proof, pp. 31, 33).

Lemma B.1 transports mass through $K+1$ overlapping balls and concludes $\mu(T,\mathcal{B}_K) \geq (1-\varepsilon)^K \mu_0(\bigcup_k \mathcal{B}_k)$. Its proof:

"Write $[0,T] = \bigcup_{k\in[1,M]}[t_{k-1}, t_k]$ where $t_k = kT/K$, and proceed by backward induction:"

$$\mu(T,\mathcal{B}_K) = \mu(T,\mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T,\mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1},\mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1-\varepsilon)\mu(t_{K-1},\mathcal{B}_{K-1}),$$

"where the last inequality follows from Lemma B.2. Using $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$ whenever $|k-k'| \geq 2$, we arrive to"

$$\mu(T,\mathcal{B}_k) \geq (1-\varepsilon)^K \Big(\sum_{k=1}^{K} \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0) \Big) = (1-\varepsilon)^K \mu_0\Big(\bigcup_{k\in[0,K]} \mathcal{B}_k \Big).$$

Our formalization. machine-checked

Stripped of the measure-theoretic bookkeeping, the backward induction is a clean arithmetic fact about a nonnegative sequence with per-step retention. We prove by induction on K that if $m \leq a_0$ and $(1-\varepsilon)\, a_k \leq a_{k+1}$ for all k, then $(1-\varepsilon)^K m \leq a_K$. The base case is $m \leq a_0$; the step chains $(1-\varepsilon)^{n+1} m = (1-\varepsilon)((1-\varepsilon)^n m) \leq (1-\varepsilon)\, a_n \leq a_{n+1}$. The key step is $0 \leq 1-\varepsilon$, which lets the induction hypothesis be scaled by $(1-\varepsilon)$ without reversing the inequality before applying the retention hypothesis.

```
/-- L9: geometric retention through a chain. With retention factor `1 - ε ∈ [0, 1]` per
↪ step and
initial mass at least `m ≥ 0`, after `K` steps at least `(1 - ε)^K · m` remains. -/
theorem ball_chain_geom {ε : ℝ} (hε1 : ε ≤ 1) (a : ℕ → ℝ) {m : ℝ}
    (h0 : m ≤ a 0) (hstep : ∀ k, (1 - ε) * a k ≤ a (k + 1)) :
    ∀ K : ℕ, (1 - ε) ^ K * m ≤ a K := by
  have h1ε : (0 : ℝ) ≤ 1 - ε := by linarith
  intro K
  induction K with
  | zero => simpa using h0
  | succ n ih =>
    calc (1 - ε) ^ (n + 1) * m
        = (1 - ε) * ((1 - ε) ^ n * m) := by ring
      _ ≤ (1 - ε) * a n := by exact mul_le_mul_of_nonneg_left ih h1ε
      _ ≤ a (n + 1) := hstep n
```

Leaves/BallChain.lean

2.7 Pigeonhole

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4 Part 1, App. B.3, p. 35).

"Part 1. There exists an open ball $\mathcal{B} \subset \operatorname{supp}\mu_0 \cup \operatorname{supp}\nu_0$ such that $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$. We now claim that there exists some $x^* \in \mathcal{B}$ such that"

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\mu_0(\mathrm{d}x) \neq \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\nu_0(\mathrm{d}x).$$

"Indeed if this were to be false, then we'd have"

$$x^* = \frac{1}{\mu_0(\mathcal{B}) - \nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,(\nu_0(\mathrm{d}x) - \mu_0(\mathrm{d}x))$$

"for all $x^* \in \mathcal{B}$, which cannot hold."

Our formalization. machine-checked

The displayed contradiction is "x^* equals a fixed vector for all $x^* \in \mathcal{B}$" — i.e. the identity would be constant on the open ball $\mathcal{B}$, impossible. We isolate exactly that general fact: in a nontrivial real normed space, an open ball of positive radius contains a point distinct from any prescribed a. Pick $u \neq 0$ and set $v = (r/2\|u\|)\,u$, so $\|v\| = r/2 < r$ and $v \neq 0$; then c and $c+v$ both lie in $B(c,r)$ (at distances 0 and $r/2$) and are distinct. If $c = a$, return $c+v$ (which differs from $c = a$ because $v \neq 0$); otherwise return c. The key step is that a positive-radius ball is never a single point, so it cannot be forced to one value — the contradiction the pigeonhole argument exploits.

```
/-- L10: an open ball of positive radius in a nontrivial real normed space contains a
 ↪  point distinct
from any prescribed `a`. Hence no map can be forced to a single value on the ball, which
 ↪  is the
contradiction the pigeonhole step exploits. -/
theorem exists_ne_in_ball (c a : E) {r : ℝ} (hr : 0 < r) :
    ∃ x ∈ Metric.ball c r, x ≠ a := by
  obtain ⟨u, hu⟩ := exists_ne (0 : E)
  have hunorm : 0 < ‖u‖ := norm_pos_iff.mpr hu
  set v : E := (r / (2 * ‖u‖)) • u with hv
  have hvnorm : ‖v‖ = r / 2 := by
    rw [hv, norm_smul, Real.norm_of_nonneg (by positivity)]
    field_simp
  have hvne : v ≠ 0 := by
    rw [← norm_pos_iff, hvnorm]; linarith
  -- `c` and `c + v` both lie in the ball and are distinct
  have hc_mem : c ∈ Metric.ball c r := Metric.mem_ball_self hr
  have hcv_mem : c + v ∈ Metric.ball c r := by
    rw [Metric.mem_ball, dist_eq_norm]
    have h1 : c + v - c = v := by abel
    rw [h1, hvnorm]; linarith
  -- at least one of the two distinct points differs from `a`
  rcases eq_or_ne c a with hca | hca
  · refine ⟨c + v, hcv_mem, fun h => hvne ?_⟩
    have h2 : c + v = c + 0 := by rw [add_zero, h, hca]
    exact add_left_cancel h2
  · exact ⟨c, hc_mem, hca⟩
```

Leaves/Pigeonhole.lean

2.8 Disjoint hulls give non-colinear barycenters

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).
The induction hypothesis (3.3) is

$$\operatorname{conv}_g \operatorname{supp} \mu_0^i \cap \operatorname{conv}_g \operatorname{supp} \mu_0^j = \emptyset \qquad \text{for } i \neq j \in [\![1, N-1]\!]. \tag{3.3}$$

The induction step then asserts, without proof:
"Since $\operatorname{supp} \mu_0^i \subset \mathbb{Q}_1^{d-1}$, (3.3) implies that"

$$\mathbb{E}_{\mu_0^i}[x] \text{ is not colinear with } \mathbb{E}_{\mu_0^j}[x] \qquad \text{for } i \neq j \in [\![1, N-1]\!].$$

Our formalization. machine-checked
Within an open hemisphere spherical geodesics are radial projections of Euclidean chords, so we model the geodesic hull as $\operatorname{conv}_g(s) = \operatorname{cone}(s) \cap \mathbb{S}^{d-1}$ — the unit vectors that are nonnegative combinations of s (`geodesicHull`). The empirical barycenter $\sum_p w_p\, p$ with $w_p \geq 0$ is a

manifest conical combination of s, and the cone is closed under nonnegative scaling, so the normalization $\|b\|^{-1}b$ of any nonzero conical point lies in $\mathrm{conv}_g(s)$. If the two barycenters were colinear — for vectors in the positive orthant "colinear" means a *positive* multiple, i.e. `SameRay` — then $b = k\,c$ with $k > 0$, so their normalizations coincide, $\|b\|^{-1}b = \|c\|^{-1}c$. That common unit direction would then sit in *both* geodesic hulls, contradicting their disjointness. The key step is that positive colinearity forces a shared normalized direction, which disjoint hulls forbid.

```lean
/-- The geodesic-convex hull of `s`, modeled within a hemisphere as `cone(s) ∩ 𝕊^{d-1}`:
 ↪  the unit
vectors that are nonnegative combinations of `s`. Faithful to `conv_g` when `s` lies in
 ↪  an open
hemisphere (e.g. the orthant `Q₁^{d-1}`), since spherical geodesics are radial
 ↪  projections of chords. -/
def geodesicHull (s : Finset (Eucl d)) : Set (Eucl d) :=
  {x | ‖x‖ = 1 ∧ inConicalSpan s x}

/-- **Core of finding F2.** If two nonzero conical points `b`, `c` (the unnormalized
 ↪  barycenters) are
colinear — `SameRay ℝ b c`, i.e. positive multiples, which is what "colinear" means for
 ↪  vectors in the
positive orthant — then their common normalized direction lies in *both* geodesic hulls.
 ↪  -/
theorem geodesicHull_inter_nonempty_of_sameRay {s₁ s₂ : Finset (Eucl d)} {b c : Eucl d}
    (hb : inConicalSpan s₁ b) (hc : inConicalSpan s₂ c) (hb0 : b ≠ 0) (hc0 : c ≠ 0)
    (hsr : SameRay ℝ b c) : (geodesicHull s₁ ∩ geodesicHull s₂).Nonempty := by
  obtain ⟨r₁, r₂, hr₁, hr₂, hrel⟩ := hsr.exists_pos hb0 hc0
  -- `b = k • c` with `k = r₁⁻¹ r₂ > 0`
  set k : ℝ := r₁⁻¹ * r₂ with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr₁), one_smul]
  -- the normalized vectors coincide
  have key : ‖b‖⁻¹ • b = ‖c‖⁻¹ • c := by
    have hnc : ‖c‖ ≠ 0 := norm_ne_zero_iff.mpr hc0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  refine ⟨‖b‖⁻¹ • b, ?_⟩
  rw [Set.mem_inter_iff]
  exact ⟨mem_geodesicHull_normalize hb hb0, key ▸ mem_geodesicHull_normalize hc hc0⟩

/-- **Finding F2, discharged.** Empirical measures (with nonnegative weights) in the
 ↪  orthant whose
geodesic hulls are disjoint have non-colinear barycenters. This is the step Proposition
 ↪  3.1's
induction asserts without proof. -/
theorem barycenter_noncolinear_of_disjoint_hull {s₁ s₂ : Finset (Eucl d)} {w₁ w₂ : Eucl
 ↪  d → ℝ}
    (hw₁ : ∀ p ∈ s₁, 0 ≤ w₁ p) (hw₂ : ∀ p ∈ s₂, 0 ≤ w₂ p)
    (hb0 : (∑ p ∈ s₁, w₁ p • p) ≠ 0) (hc0 : (∑ p ∈ s₂, w₂ p • p) ≠ 0)
    (hdisj : Disjoint (geodesicHull s₁) (geodesicHull s₂)) :
    ¬ SameRay ℝ (∑ p ∈ s₁, w₁ p • p) (∑ p ∈ s₂, w₂ p • p) := by
  intro hsr
  obtain ⟨u, hu⟩ := geodesicHull_inter_nonempty_of_sameRay
    (barycenter_mem_conicalSpan hw₁) (barycenter_mem_conicalSpan hw₂) hb0 hc0 hsr
  rw [Set.mem_inter_iff] at hu
  exact Set.disjoint_left.mp hdisj hu.1 hu.2
```

Leaves/BarycenterNonColinear.lean

Strengthening (finding F2). Proposition 3.1's induction *asserts* "(3.3) [disjoint geodesic hulls] $\Rightarrow \mathbb{E}_{\mu_0^i}[x]$ not colinear with $\mathbb{E}_{\mu_0^j}[x]$" with no justification — the one genuine rigor gap the adversarial review flagged. This leaf supplies the kernel-checked proof for the empirical regime (the Dirac targets of Theorem 1.1 and the empirical targets of the restricted Theorem 1.2), under the faithful hemisphere model $\mathrm{conv}_g(s) = \mathrm{cone}(s) \cap \mathbb{S}^{d-1}$.

2.9 F2, general probability measures

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1 induction step, pp. 16-17).
The same unproved assertion as above, now read for arbitrary $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ with barycenter $\mathbb{E}_{\mu_0}[x] = \int x\, \mu_0(\mathrm{d}x)$:

$$\mathrm{conv}_g \operatorname{supp} \mu_0 \cap \mathrm{conv}_g \operatorname{supp} \nu_0 = \emptyset \implies \mathbb{E}_{\mu_0}[x] \text{ not colinear with } \mathbb{E}_{\nu_0}[x].$$

Our formalization. machine-checked
The finite case above is elementary; for a general probability measure the only extra ingredient is the standard fact that the barycenter $\int x\, \mathrm{d}\mu$ lies in any closed convex set carrying full mass — Mathlib's `Convex.integral_mem`. Wrapping it as `barycenter_mem_of_supportedIn` (from $\mu(s^c) = 0$ to $\int x\, \mathrm{d}\mu \in s$), the same `SameRay` argument runs verbatim: both normalized barycenters land in their respective closed convex *cones* s_1, s_2, the shared unit direction sits in both, and disjointness on the sphere is contradicted. The key step is that `Convex.integral_mem` is the measure-theoretic analogue of "the barycenter lies in the convex hull of the support," so no optimal-transport axiom is reintroduced.

```
/-- The barycenter of a probability measure lies in any closed convex set carrying full
 ↪  mass
(`μ Sᶜ = 0`). This is the standard measure-theoretic fact the general case of finding F2
 ↪  needs; it
rests only on Mathlib's `Convex.integral_mem`, not on the optimal-transport axioms. -/
theorem barycenter_mem_of_supportedIn {μ : Measure (Eucl d)} [IsProbabilityMeasure μ]
    (hint : Integrable (fun x : Eucl d ⇒ x) μ) {s : Set (Eucl d)}
    (hconv : Convex ℝ s) (hclosed : IsClosed s) (hsupp : μ sᶜ = 0) :
    barycenter μ ∈ s := by
  have hae : ∀ᵐ x ∂μ, x ∈ s := by
    rw [ae_iff]; exact hsupp
  rw [barycenter]
  exact hconv.integral_mem hclosed hae hint

/-- **Finding F2, general case.** Two probability measures whose supports lie in
 ↪  disjoint closed
convex cones `s₁`, `s₂` (disjoint on the sphere) have non-colinear barycenters.
 ↪  Generalizes
`barycenter_noncolinear_of_disjoint_hull` from empirical to arbitrary probability
 ↪  measures; kernel-
clean, via `barycenter_mem_of_supportedIn`. -/
theorem barycenter_noncolinear_of_disjoint_hull_general
    {μ ν : Measure (Eucl d)} [IsProbabilityMeasure μ] [IsProbabilityMeasure ν]
    (hμi : Integrable (fun x : Eucl d ⇒ x) μ) (hνi : Integrable (fun x : Eucl d ⇒ x) ν)
    {s₁ s₂ : Set (Eucl d)}
    (hc₁ : Convex ℝ s₁) (hcl₁ : IsClosed s₁) (hcone₁ : ∀ x ∈ s₁, ∀ a : ℝ, 0 ≤ a → a • x ∈
 ↪  s₁)
    (hc₂ : Convex ℝ s₂) (hcl₂ : IsClosed s₂) (hcone₂ : ∀ x ∈ s₂, ∀ a : ℝ, 0 ≤ a → a • x ∈
 ↪  s₂)
    (hμs : μ s₁ᶜ = 0) (hνs : ν s₂ᶜ = 0)
    (hb0 : barycenter μ ≠ 0) (hc0 : barycenter ν ≠ 0)
    (hdisj : ∀ u : Eucl d, ‖u‖ = 1 → u ∈ s₁ → u ∉ s₂) :
    ¬ SameRay ℝ (barycenter μ) (barycenter ν) := by
  intro hsr
  set b := barycenter μ with hbdef
  set c := barycenter ν with hcdef
  obtain ⟨r₁, r₂, hr₁, hr₂, hrel⟩ := hsr.exists_pos hb0 hc0
  have hbs₁ : b ∈ s₁ := barycenter_mem_of_supportedIn hμi hc₁ hcl₁ hμs
  have hcs₂ : c ∈ s₂ := barycenter_mem_of_supportedIn hνi hc₂ hcl₂ hνs
  set k : ℝ := r₁⁻¹ * r₂ with hkdef
  have hk : 0 < k := by positivity
  have hbk : b = k • c := by
    rw [hkdef, mul_smul, ← hrel, smul_smul, inv_mul_cancel₀ (ne_of_gt hr₁), one_smul]
  have key : ‖b‖⁻¹ • b = ‖c‖⁻¹ • c := by
    have hnc : ‖c‖ ≠ 0 := norm_ne_zero_iff.mpr hc0
    rw [hbk, norm_smul, Real.norm_eq_abs, abs_of_pos hk, smul_smul]
    congr 1
    field_simp
  set u := ‖b‖⁻¹ • b with hu
  have hunorm : ‖u‖ = 1 := by
    rw [hu, norm_smul, norm_inv, norm_norm, inv_mul_cancel₀ (norm_ne_zero_iff.mpr hb0)]
  have hus₁ : u ∈ s₁ := by rw [hu]; exact hcone₁ b hbs₁ _ (by positivity)
  have hus₂ : u ∈ s₂ := by rw [key]; exact hcone₂ c hcs₂ _ (by positivity)
  exact hdisj u hunorm hus₁ hus₂
```

Leaves/BarycenterNonColinear.lean

Strengthening (finding F2, residual). The finite leaf above closes F2 for the empirical regime; this companion closes the remaining general-measure case, extending the non-colinearity to arbitrary probability measures supported in disjoint closed convex cones. It rests only on Mathlib's `Convex.integral_mem` and introduces **no new axioms** — in particular it does not reach for the optimal-transport (W_2) axiom.

2.10 Linearized OT bound

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.2, p. 24).
"Lemma 5.2. Suppose $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and $T^1, T^2 : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ measurable, with T^1 bijective. Then"

$$\mathsf{W}_2(T^1_\# \mu, T^2_\# \mu) \lesssim \|T^1 - T^2\|_{L^2(\mu)}. \tag{5.2}$$

"Proof of Lemma 5.2. Since T^1 is bijective, there is a measurable $\psi : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that $\psi(T^1(x)) = T^2(x)$ for all $x \in \mathbb{S}^{d-1}$. Then"

$$\mathsf{W}_2^2(T^1_\# \mu, T^2_\# \mu) \lesssim \int \|x - \psi(x)\|^2 (T^1_\# \mu)(\mathrm{d}x) = \|T^1 - T^2\|^2_{L^2(\mu)}.$$

Our formalization. axiomatised
This is the map-induced coupling bound. Since T^1 is bijective there is ψ with $\psi \circ T^1 = T^2$, and the deterministic coupling $(\mathrm{id}, \psi)_\# (T^1_\# \mu)$ between the two pushforwards realizes transport cost $\int \|x - \psi(x)\|^2 \, \mathrm{d}(T^1_\# \mu) = \|T^1 - T^2\|^2_{L^2(\mu)}$, bounding W_2^2. Mathlib `v4.31.0` has no W_2, so the Lean statement is asserted over the labeled optimal-transport axiom `W2_map_le_L2`; the node is therefore **axiomatised**, not a kernel proof. Its effective status is the minimum over its dependency closure, which contains the W_2 axiom.

```
/-- L7 (Lemma 5.2): the `W₂` distance between two pushforwards of `μ` is bounded by the
↪  `L²(μ)`
distance of the maps. Rests on the optimal-transport axiom `W2_map_le_L2`. -/
theorem lemma_5_2 (μ : Measure (Eucl d)) (T₁ T₂ : Eucl d → Eucl d) :
    W2 (μ.map T₁) (μ.map T₂) ≤ Real.sqrt (∫ x, ‖T₁ x - T₂ x‖ ^ 2 ∂μ) :=
  W2_map_le_L2 μ T₁ T₂
```

Leaves/Coupling.lean

2.11 Markov bound

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Claim 2, App. B.5, p. 39). "Claim 2. Suppose $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$ and $x_0 \in \mathbb{S}^{d-1}$ satisfy $\mathsf{W}_2(\mu, \delta_{x_0}) \leq \eta_2$. There exists a universal constant $C > 0$ such that $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ for all $\eta_3 > 0$."
"Proof of Claim 2. By compactness of $\mathbb{S}^{d-1}$ and Kantorovich-Rubinstein duality,"

$$\mathsf{W}_1(\mu, \delta_{x_0}) = \sup_{\mathrm{Lip}(g) \leq 1} \int g\,(\mu - \delta_{x_0}) \leq C \cdot \eta_2$$

"for some numerical constant $C > 0$. Hence, for $g : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ defined as"

$$g(x) = \begin{cases} 1 & x \in B(x, \eta_3) \\ 1 - \dfrac{1-\eta_3}{\eta_3} & x \in B(x, \eta_3) \cap B(x, 2\eta_3) \\ 0 & x \notin B(x, 2\eta_3), \end{cases}$$

"we obtain $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$."

Our formalization. axiomatised
We take the cleaner truncated-distance bump $f(x) = \min(\eta_3,\ d(x, x_0))$ in place of the paper's hat. It is **machine-checked 1-Lipschitz** — `distBump_lipschitz`, obtained by truncating the 1-Lipschitz distance through the map $t \mapsto \min(\eta_3, t)$ and routed through the `dist`-characterization to dodge the `EuclideanSpace/PiLp` extended-metric instance diamond — vanishes at x_0, and dominates $\eta_3 \cdot \mathbf{1}_{\{d(\cdot,x_0) \geq \eta_3\}}$. Hence $\eta_3\, \mu\{d(\cdot, x_0) \geq \eta_3\} \leq \int f \,\mathrm{d}\mu = \int f \,\mathrm{d}\mu - f(x_0) \leq \mathsf{W}_1(\mu, \delta_{x_0}) \leq C\eta_2$, and since $\{\eta_3 \leq d(\cdot, x_0)\} = B(x_0, \eta_3)^c$ this is $1 - \mu(B(x_0, \eta_3)) \leq C\eta_2/\eta_3$ for a probability measure. The Lipschitz bound is kernel-checked, but the middle inequality is Kantorovich-Rubinstein duality, supplied here by the labeled W_1 axiom `W1_ge_of_lipschitz`; so the node as a whole is **axiomatised**.

```
/-- The truncated-distance bump `f(x) = min(η₃, d(x, x₀))`. -/
noncomputable def distBump (x₀ : Eucl d) (η₃ : ℝ) (x : Eucl d) : ℝ := min η₃ (dist x x₀)

/-- The bump is `1`-Lipschitz (a truncation of the `1`-Lipschitz distance). Proved
 ↪  through the
`dist`-characterization to avoid the `EuclideanSpace`/`PiLp` extended-metric instance
 ↪  diamond. -/
theorem distBump_lipschitz (x₀ : Eucl d) (η₃ : ℝ) : LipschitzWith 1 (distBump x₀ η₃) :=
 ↪  by
  rw [lipschitzWith_iff_dist_le_mul]
  intro x y
  -- `t ↦ min η₃ t` is 1-Lipschitz on ℝ (no PiLp instance diamond on the codomain)
  have hL : LipschitzWith 1 (fun t : ℝ ⇒ min η₃ t) := LipschitzWith.const_min
 ↪  LipschitzWith.id η₃
  have hmin := hL.dist_le_mul (dist x x₀) (dist y x₀)
  simp only [distBump, Real.dist_eq, NNReal.coe_one, one_mul] at hmin ⊢
  exact hmin.trans (abs_dist_sub_le x y x₀)

/-- **Lemma L8 / Claim 2 (Markov bound).** For a probability measure with
`W₁(μ, δ_{x₀}) ≤ C η₂`, the mass at geodesic/Euclidean distance `≥ η₃` from `x₀` is at
 ↪  most
`C η₂ / η₃`. Equivalently `1 - μ(B(x₀, η₃)) ≤ C η₂ / η₃`. Rests on the `W₁` axiom (KR
 ↪  duality):
status `math.axiomatised`. -/
theorem markov_bound (μ : Measure (Eucl d)) [IsProbabilityMeasure μ]
    (x₀ : Eucl d) (η₂ η₃ C : ℝ) (hη₃ : 0 < η₃)
    (hW1 : W1 μ (Measure.dirac x₀) ≤ C * η₂) :
    μ.real {x | η₃ ≤ dist x x₀} ≤ C * η₂ / η₃ := by
  set s : Set (Eucl d) := {x | η₃ ≤ dist x x₀} with hs
  have hcont : Continuous (fun x : Eucl d ⇒ dist x x₀) := by fun_prop
  have hsmeas : MeasurableSet s := measurableSet_le measurable_const hcont.measurable
  -- the bump is integrable (bounded by η₃ on a finite measure)
  have hbump_int : Integrable (distBump x₀ η₃) μ := by
    refine Integrable.mono' (integrable_const η₃)
      (distBump_lipschitz x₀ η₃).continuous.aestronglyMeasurable ?_
    filter_upwards with x
    rw [Real.norm_eq_abs, abs_of_nonneg (distBump_nonneg hη₃.le x)]
    exact min_le_left _ _
  have hind_int : Integrable (s.indicator (fun _ ⇒ η₃)) μ :=
    (integrable_const η₃).indicator hsmeas
  -- η₃ · 𝟙_s ≤ bump pointwise
  have hdom : s.indicator (fun _ ⇒ η₃) ≤ distBump x₀ η₃ := by
    intro x
    by_cases hx : x ∈ s
    · rw [Set.indicator_apply, if_pos hx]; exact (distBump_eq_outside hx).ge
    · rw [Set.indicator_apply, if_neg hx]; exact distBump_nonneg hη₃.le x
  -- ∫ η₃·𝟙_s dμ = η₃ · μ(s)
  have hint_ind : ∫ x, s.indicator (fun _ ⇒ η₃) x ∂μ = μ.real s * η₃ := by
    rw [integral_indicator_const η₃ hsmeas, smul_eq_mul]
  -- ∫ bump dδ_{x₀} = f(x₀) = 0
  have hdirac : ∫ x, distBump x₀ η₃ x ∂(Measure.dirac x₀) = 0 := by
    rw [integral_dirac]
    simp [distBump, min_eq_right hη₃.le]
  -- KR duality (axiom): ∫ bump dμ - ∫ bump dδ ≤ W₁
  have hdual := W1_ge_of_lipschitz μ (Measure.dirac x₀) (distBump x₀ η₃)
 ↪  (distBump_lipschitz x₀ η₃)
  rw [hdirac, sub_zero] at hdual
  -- assemble: η₃ · μ(s) ≤ ∫ bump dμ ≤ W₁ ≤ C η₂
  have hmono : μ.real s * η₃ ≤ ∫ x, distBump x₀ η₃ x ∂μ := by
```

```
Leaves/MarkovBound.lean
```

3 Mass transport and clustering

The ReLU-gated drift $\mathbf{W}(\mathbf{U}x+b)_+$ transports mass through a chain of overlapping balls, retaining a $(1-\varepsilon)^K$ fraction of it after K parameter switches. Self-attention then clusters a measure supported in a single hemisphere to a point, and chaining the two steers an ensemble of disjoint-support atomless measures to prescribed discrete targets.

3.1 Single overlapping ball pair (Lemma B.2)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.2, p. 31). Consider two open balls $\mathcal{B}_0, \mathcal{B}_1 \subset \mathbb{S}^{d-1}$ such that $\mathcal{B}_0 \cap \mathcal{B}_1 \neq \emptyset$. For any $\varepsilon > 0$ and $T > 0$, there exist $\mathbf{W}, \mathbf{U} \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ such that for any $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$, the unique solution μ to (B.1) satisfies

$$\mu(T, \mathcal{B}_0 \cap \mathcal{B}_1) \geq (1-\varepsilon)\, \mu_0(\mathcal{B}_0). \tag{B.3}$$

Moreover $\mu(T) = \phi^T_{\#}\mu_0$ where the Lipschitz-continuous and invertible flow map $\Phi^t : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ of (4.1) satisfies, for all $t \in [0, T]$,

$$(\phi^t)_{|\mathbb{S}^{d-1}\setminus\mathcal{B}_0} \equiv \mathrm{Id}\,.$$

Furthermore, for any fixed $\omega \in \operatorname{int}\mathcal{B}_0$ we can choose $\mathbf{W}, \mathbf{U}$ and b so that the solution to (B.1) satisfies $W_2(\mu(T), \alpha) \leq \varepsilon$ where $\alpha(A) = \mu_0(\mathcal{B}_0)\,\delta_\omega(A) + \mu_0(A \setminus \mathcal{B}_0)$, for any Borel $A \subset \mathbb{S}^{d-1}$.
Proof sketch (full proof p. 32). Let z and R be the center and radius of $\mathcal{B}_0$ and fix $\omega \in \operatorname{int}(\mathcal{B}_0 \cap \mathcal{B}_1)$. The authors choose $\mathbf{U}, b$ and a $\mathbf{W}$ with $\mathbf{W}\mathbf{1} = \omega$ so that the gated drift is $(-\cos d_g(z,x) + \cos R)_+\,\omega = (\cos R - \langle z, x\rangle)_+\,\omega$, and record (B.4) $(\cos R - \langle z, x\rangle)_+ > 0 \iff x \in \mathcal{B}_0$ (see the strengthening below). Then (B.5)

$$\tfrac{d}{dt}\langle x(t), \omega\rangle = (\cos R - \langle z, x(t)\rangle)_+(1 - \langle x(t), \omega\rangle^2),$$

which is positive whenever $x(t) \in \mathcal{B}_0 \setminus \{\omega\}$. Hence for small $\delta > 0$ there is a time T_δ with $\phi^{T_\delta}(x) \in \mathcal{B}_0 \cap \mathcal{B}_1$ for $x \in B(z, R-\delta)$ (B.7); pushing forward, $\mu(T_\delta, \mathcal{B}_0 \cap \mathcal{B}_1) = \mu_0((\phi^{T_\delta})^{-1}(\mathcal{B}_0 \cap \mathcal{B}_1)) \geq \mu_0(B(z, R-\delta))$ (B.8), and taking δ small enough that $\mu_0(B(z, R-\delta)) \geq (1-\varepsilon)\mu_0(\mathcal{B}_0)$ gives (B.3) after rescaling time. The gate vanishes off $\mathcal{B}_0$, so $\phi^t \equiv \mathrm{Id}$ there. The bound to α follows by taking $\mathcal{B}_1 = B(\omega, \eta) \subset \mathcal{B}_0$ and a W_1 Kantorovich-Rubinstein estimate, sending η and the slack to zero.

What we strengthen. As printed the proof takes $\mathbf{U} = -\mathbf{1}z^\top$, $b = \cos(R)\mathbf{1}$ (with $\mathbf{W}\mathbf{1} = \omega$), so $\mathbf{U}x = -\langle z, x\rangle\mathbf{1}$, giving the gate coefficient $(\cos R - \langle z, x\rangle)_+$, and asserts in (B.4) that this is positive iff $x \in \mathcal{B}_0 = B(z, R)$. But

$$\cos R - \langle z, x\rangle > 0 \iff \langle z, x\rangle < \cos R \iff d_g(z, x) > R \iff x \notin \mathcal{B}_0,$$

so as written the gate is active on the **complement** of $\mathcal{B}_0$: the interior mass it is meant to move sees a zero gate, and the lemma transports nothing. The fix is a one-sign typo — take $\mathbf{U} = +\mathbf{1}z^\top$, $b = -\cos(R)\mathbf{1}$, giving the gate $(\langle z, x\rangle - \cos R)_+$, positive exactly on $\mathcal{B}_0$ — after which (B.3)-(B.5) and the rest of the argument go through and Lemma B.2 holds as stated. The corrected equivalence $g(x) > 0 \iff d_g(z, x) > R$ is the kernel-checked leaf **L2** (`gate_pos_iff_dist`, machine-checked); the same inversion is corroborated independently by Proposition 4.2, Step 3 (which states the identical construction with the correct activation side) and by experiment E1. This is finding **F1**; see `ERRATA.md`.

Two further hypotheses turned out to be load-bearing and are restored in the formal statement (finding **F12** and the M4 discharge review): the measure must be a **probability measure** (an infinite measure can stack unbounded mass arbitrarily close to the cap rim, where any single gated block moves it arbitrarily slowly, refuting the uniform $(1-\varepsilon)$ retention), and the caps must be **sub-hemisphere** ($R_i < \pi/2$: a larger cap can contain both ω and $-\omega$, where every field of the form $g\,\mathbf{P}_x^\perp\omega$ vanishes, so mass near $-\omega$ stalls). The paper's printed quantifier order — parameters first, "for any μ_0" second — is also refutable (a Dirac slid to the rim defeats any fixed gate); the formal statement quantifies $\forall\mu\,\exists\theta$, which is the order the paper's own proof establishes. That is erratum candidate **E3** in `ERRATA.md`.

Our formalization. machine-checked

The authors' strategy is the gated-drift construction above: a single ReLU gate, active on $\mathcal{B}_0$ (after the F1 sign correction), pushes the interior mass of $\mathcal{B}_0$ along $+\omega$ into $\mathcal{B}_0 \cap \mathcal{B}_1$ while leaving everything off $\mathcal{B}_0$ fixed, so a $(1-\varepsilon)$ fraction is retained in the overlap. This is now a **proved theorem**: `#print axioms lemma_B_2` lists only `propext`, `Classical.choice`, `Quot.sound`.

The proof *recenters* the construction at a point ω of the open overlap instead of analyzing an off-center gate: the geodesic triangle inequality (Mathlib's `angle_le_angle_add_angle`) places the shrunken cap $B(z_0, R_0 - \delta)$ inside a cap around ω and a small target cap around ω inside $\mathcal{B}_0 \cap \mathcal{B}_1$, so the already-verified self-centered machinery applies verbatim. An **amplitude-scaled gated block** (`scaledGatedBlock`) realizes the paper's $\|\theta\| \sim C/(T\varepsilon)$ parameter freedom, buying the logistic reaching budget at any fixed horizon T with one switch. The chain, all machine-checked:

- `gatedBlock` / `scaledGatedBlock` — the B.5 field $A\,\chi(\|x\|)(\langle z, x\rangle - \cos R)_+\,\mathbf{P}_x^\perp\omega$ as a genuine well-posed `Block` (globally Lipschitz via a *vanishing-gluing* lemma, bounded, radially tangent);
- `hasDerivAt_inner_gatedFlow` — the logistic gate ODE $u' = g\,(1-u^2)$ (B.5) along the flow;
- `logistic_flow_reach` — the finite-time reaching estimate (B.7) via the log-odds substitution;
- `exists_closed_sublevel_mass_ge` — the shrunken-cap mass bound (B.6);
- `le_measureFlow_of_mapsTo` — the B.8 pushforward step turning the point-set contraction into mass retention in $\mathcal{B}_0 \cap \mathcal{B}_1$.

```
/-- **Lemma B.2** (single ball pair). Mass in the geodesic ball `B(z₀,R₀)` is pushed into
`B(z₀,R₀) ∩ B(z₁,R₁)`, retaining `(1-ε)`, with one switch. PROVED
 ↪  (`math.machine-checked`)
by recentring the self-centered gated flow at a point of the overlap. (Finding F1: the
gate sign is `U = +z 𝟙ᵀ, b = -cos(R) 𝟙`; see `ERRATA.md`.) -/
theorem lemma_B_2 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (_hd : 2 ≤ d)
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (z₀ z₁ : Eucl d) (hz₀ : z₀ ∈ sphere d) (hz₁ : z₁ ∈ sphere d) (R₀ R₁ : ℝ)
    (hR₀ : R₀ ∈ Set.Ioo 0 (Real.pi / 2)) (hR₁ : R₁ ∈ Set.Ioo 0 (Real.pi / 2))
    (hcap : (geodesicBall z₀ R₀ ∩ geodesicBall z₁ R₁).Nonempty) :
    ∃ θ : Params d, switches θ ≤ 1 ∧
      (1 - ENNReal.ofReal ε) * μ (geodesicBall z₀ R₀) ≤
        (measureFlow θ T μ) (geodesicBall z₀ R₀ ∩ geodesicBall z₁ R₁)
```

Statements/MidLevel.lean

3.2 Chain of overlapping balls (Lemma B.1)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma B.1, p. 31).
Consider $K+1$ open balls $\mathcal{B}_K, \dots, \mathcal{B}_1, \mathcal{B}_0 \subset \mathbb{S}^{d-1}$ satisfying

$$\mathcal{B}_k \cap \mathcal{B}_{k-1} \neq \emptyset \quad \text{for } k \in [\![1, K]\!], \qquad \mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset \quad \text{if } |k-k'| \geq 2.$$

Then for any $T > 0$ and $\varepsilon > 0$, there exist $(\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, piecewise constant having at most K switches, such that for any $\mu_0 \in \mathcal{P}(\mathbb{S}^{d-1})$, the corresponding unique solution μ to

$$\begin{cases} \partial_t \mu(t) + \operatorname{div}(\mathbf{P}_x^\perp \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+ \mu(t)) = 0 & \text{on } [0,T] \times \mathbb{S}^{d-1}, \\ \mu(0) = \mu_0 & \text{on } \mathbb{S}^{d-1}, \end{cases} \tag{B.1}$$

satisfies

$$\mu(T, \mathcal{B}_K) \geq (1-\varepsilon)^K \mu_0\Big(\bigcup_k \mathcal{B}_k\Big).$$

Moreover, $\mu(T) = \phi^T_\# \mu_0$ for a Lipschitz-continuous, invertible map $\phi^t : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ which satisfies, for all $t \in [0, T]$,

$$\phi^t(x) = x \quad \text{for } x \notin \bigcup_k \mathcal{B}_k. \tag{B.2}$$

Proof sketch (full proof p. 33). Backward induction on the chain. Write $[0, T) = \bigcup_{k\in[\![1,K]\!]}[t_{k-1}, t_k)$ with $t_k = kT/K$ and apply Lemma B.2 to the adjacent pair $(\mathcal{B}_{k-1}, \mathcal{B}_k)$ on each subinterval. Then

$$\mu(T, \mathcal{B}_K) = \mu(T, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + \mu(T, \mathcal{B}_K \cap \mathcal{B}_{K-1}) \geq \mu(t_{K-1}, \mathcal{B}_K \setminus \mathcal{B}_{K-1}) + (1-\varepsilon)\, \mu(t_{K-1}, \mathcal{B}_{K-1}),$$

the last step by Lemma B.2. The non-adjacency disjointness $\mathcal{B}_k \cap \mathcal{B}_{k'} = \emptyset$ for $|k-k'| \geq 2$ keeps the steps from interfering, and iterating gives

$$\mu(T, \mathcal{B}_K) \geq (1-\varepsilon)^K \Big(\sum_{k=1}^K \mu_0(\mathcal{B}_k \setminus \mathcal{B}_{k-1}) + \mu_0(\mathcal{B}_0)\Big) = (1-\varepsilon)^K \mu_0\Big(\bigcup_{k\in[\![0,K]\!]} \mathcal{B}_k\Big).$$

Our formalization. machine-checked
The authors' argument is a backward induction that applies the single-pair lemma (Lemma B.2) once per adjacent overlap, each step retaining a factor $(1-\varepsilon)$. We **prove** this in Lean by induction on the chain length K: the base case is the identity schedule (`idParams`, zero switches); each step composes one `lemma_B_2` transport via `comp`, carrying the previous mass forward with `measureFlow_comp`, passing from $\mathcal{B}_k \cap \mathcal{B}_{k+1}$ to $\mathcal{B}_{k+1}$ by `measure_mono`, and adding the switch budget with `switches_comp`. With Lemma B.2 itself now discharged, **the whole Appendix-B chain is machine-checked end to end**: `#print axioms lemma_B_1` lists only `propext`, `Classical.choice`, `Quot.sound`.
Corrections over the original type-correct stub. The retained fraction multiplies $\mu(\mathcal{B}_0)$ — the mass that *starts* in the first ball and is funneled along the chain — not $\mu(\bigcup_k \mathcal{B}_k)$ (which would make the $K=0$ base case false, since $\mathcal{B}_0 \subseteq \bigcup_k \mathcal{B}_k$). The chain is a sequence of genuine sub-hemisphere geodesic balls $B(z_k, R_k)$, $R_k < \pi/2$, over a probability measure, matching the faithful `lemma_B_2` signature. The chain-overlap hypothesis `hchain` and the per-step `switches ≤ 1` budget are what make the bound hold.

```
/-- **Lemma B.1** (ball-chain retention). For a chain of `K+1` overlapping geodesic
 ↪ balls, `K`
switches retain `(1-ε)^K` of the mass initially in `B(z₀,R₀)` into the last ball. PROVED
(`math.machine-checked`) by induction on `K` over the discharged `lemma_B_2`. -/
theorem lemma_B_1 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (hd : 2 ≤ d)
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (K : ℕ) (z : ℕ → Eucl d) (hz : ∀ k, z k ∈ sphere d) (R : ℕ → ℝ)
    (hR : ∀ k, R k ∈ Set.Ioo 0 (Real.pi / 2))
    (hchain : ∀ k, (geodesicBall (z k) (R k) ∩ geodesicBall (z (k + 1)) (R (k +
 ↪  1))).Nonempty) :
    ∃ θ : Params d, switches θ ≤ K ∧
      (1 - ENNReal.ofReal ε) ^ K * μ (geodesicBall (z 0) (R 0)) ≤
        (measureFlow θ T μ) (geodesicBall (z K) (R K)) := by
  induction K with
  | zero => exact ⟨idParams d, by simp [switches_id], by simp [measureFlow_id]⟩
  | succ k ih => /- compose one lemma_B_2 step onto the chain -/
```

Statements/MidLevel.lean

3.3 Clustering a single-hemisphere measure to a point (Proposition 2.1)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.1, p. 11). Suppose $\mathbf{B} \in \mathscr{M}_{d\times d}(\mathbb{R})$ and $\operatorname{supp}\mu_0$ is contained in an open hemisphere. Then the solution μ to (1.3)-(1.2) with data μ_0 and $(\mathbf{V}(\cdot), \mathbf{B}(\cdot), \mathbf{W}(\cdot)) \equiv (I_d, \mathbf{B}, 0)$ satisfies $\operatorname{diam}(\operatorname{conv}_g \operatorname{supp}\mu(t)) \to 0$ as $t \to +\infty$. Moreover, for $\varepsilon > 0$ there exist $z \in \operatorname{conv}_g \operatorname{supp}\mu_0$ and $T > 0$ such that

$$W_\infty(\mu(T), \delta_z) \le \varepsilon \quad \text{and} \quad \inf\{t \ge 0 : W_2(\mu(t), \delta_z) \le \varepsilon\} = O\Big(\log \tfrac{1}{\varepsilon}\Big).$$

Proof sketch (full proof p. 11; an adaptation of [GLPR25, Lemma 6.4]). The characteristic flow is $\dot{x}(t) = \mathbf{P}^\perp_{x(t)} \mathscr{A}_\mathbf{B}[\mu(t)](x(t))$. Since $\operatorname{supp}\mu_0$ lies in an open hemisphere, the normalized attention vector $\gamma(x) := \mathscr{A}_\mathbf{B}[\mu_0](x)/\|\mathscr{A}_\mathbf{B}[\mu_0](x)\|$ is well defined on $\operatorname{conv}_g \operatorname{supp}\mu_0$ and points strictly into its interior. A first-order expansion at a boundary point x_0 gives $\langle x(\tau) - x_0, \gamma(x_0)\rangle > 0$ for small $\tau > 0$, so boundary points move inward and the geodesic convex hulls are strictly nested in time, $\operatorname{conv}_g \operatorname{supp}\mu(t_2) \subset \operatorname{int}\operatorname{conv}_g \operatorname{supp}\mu(t_1)$ for $t_1 < t_2$. Hence $\phi(t) := \operatorname{diam}(\operatorname{conv}_g \operatorname{supp}\mu(t))$ decreases to some $\ell \ge 0$; $\ell > 0$ would, by compactness, produce boundary points that fail to move inward, contradicting the strict interior-pointing, so $\ell = 0$. Nestedness and vanishing diameter yield a unique z with $\mu \rightharpoonup \delta_z$. Picking T with $\operatorname{diam}(\operatorname{conv}_g \operatorname{supp}\mu(T)) \le \varepsilon$ and the coupling $(\mathrm{Id}, \mathsf{T})_\# \mu(T)$, $\mathsf{T} \equiv z$, gives $W_p(\mu(T), \delta_z) \le \varepsilon$; the logarithmic rate follows from [CLPR25, Theorem 2.3].

Our formalization. axiomatised
The authors' strategy is a Lyapunov/LaSalle argument on the geodesic convex hull: with the pure self-attention drift ($\mathbf{W} = 0$, $\mathbf{V} = I_d$) the hull's diameter is strictly decreasing while its mass stays inside it, so the measure contracts to a single Dirac, the exponential contraction giving the $O(\log \frac{1}{\varepsilon})$ time. Because the drift is genuinely **measure-dependent** (self-attention

averages against the evolving measure), the statement lives on the mean-field attention layer (`AttnSchedule`, `attnMeasureFlow`) introduced by the F14 restatement — a statement over the measure-independent linear flow would be about the wrong dynamics. It is a labeled `axiom`: the diameter monotonicity and the LaSalle/Hartman-Grobman convergence are the milestone-M6 build, on top of the single mean-field well-posedness axiom (`exists_meanFieldFlow`). The clustering itself carries a one-piece (`switches θ ≤ 1`) budget, the paper's single constant parameter $(\mathbf{V}, \mathbf{B}, \mathbf{W}) \equiv (I_d, \mathbf{B}, 0)$. The conclusion is validated numerically by experiment **E2**.

```
/-- **Proposition 2.1** (clustering to a point). A sphere-supported probability measure
 ↪  in an open
hemisphere can be driven arbitrarily `W₂`-close to a Dirac mass at a sphere point, with
 ↪  one
constant attention piece. AXIOM (`math.axiomatised`): the LaSalle/Hartman-Grobman
 ↪  convergence
is the M6 build over the mean-field flow. -/
axiom prop_2_1 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (T ε : ℝ) (hT : 0 < T)
 ↪  (hε : 0 < ε)
    (e : Eucl d) (he : ‖e‖ = 1)
    (hμs : supportedIn μ (sphere d)) (hhemi : supportedIn μ {x | 0 < ⟪e, x⟫}) :
    ∃ (θ : AttnSchedule d) (z : Eucl d), AttnSchedule.durationSum θ = T ∧
      AttnSchedule.switches θ ≤ 1 ∧ z ∈ sphere d ∧
      Axioms.W2 (attnMeasureFlow θ μ) (Measure.dirac z) ≤ ε
```

Statements/MidLevel.lean

3.4 Clustering disjoint-support measures to atoms (Proposition 2.2)

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 2.2, pp. 11-12). Suppose μ_0^i has no atoms and $\mathrm{conv}_g \operatorname{supp} \mu_0^i \cap \mathrm{conv}_g \operatorname{supp} \mu_0^j = \emptyset$ for $i \neq j$. Fix $M \geq 1$, and for any $i \in [\![1, N]\!]$ consider

$$\mu_1^i := \sum_{k=1}^{M} \alpha_k^i \, \delta_{x_k^i} \in \mathscr{P}(\mathbb{S}^{d-1})$$

where $x_k^i \in \mathrm{conv}_g \operatorname{supp} \mu_0^i$, with $x_k^i = x_{k'}^j$ if and only if $(k, i) = (k', j)$. Then for any $T > 0$ and $\varepsilon > 0$ there exist piecewise constant $(\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathscr{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$ such that for any i, the corresponding solution μ^i to (1.3)-(1.2) with data μ_0^i, $\mathbf{V} \equiv 0$ and the above parameters, satisfies $W_2(\mu^i(T), \mu_1^i) \leq \varepsilon$ as well as, for $i \neq j$,

$$\mathrm{conv}_g \operatorname{supp} \mu^i(T) \cap \mathrm{conv}_g \operatorname{supp} \mu^j(T) = \emptyset.$$

The number of switches in $(\mathbf{W}, \mathbf{V}, b)$ can also be accounted for — see Remark 2.3.
Proof sketch (full proof from p. 12). Split $[0, T] = \bigcup_{i \in [\![1,N]\!]} [T_{i-1}, T_i]$ and apply Lemma B.1 separately within each interval, acting on one measure at a time. Property (B.2) — the flow is the identity off the active balls — ensures that while the construction acts on the i-th measure the others are untouched, $\mu^i(T_{i-1}) = \mu_0^i$. For a fixed i, partition the support $\mathscr{C}^i = \operatorname{supp} \mu_0^i$ into M pairwise disjoint pieces $\mathscr{C}_k^i$ with connected interiors, $\mu_0^i(\mathscr{C}_k^i) = \alpha_k^i$ and $x_k^i \in \operatorname{int} \mathscr{C}_k^i$; then

steer the mass of each piece, through a chain of overlapping balls, into a small ball around x_k^i, so $W_2(\mu^i(T), \mu_1^i) \le \varepsilon$ while the disjoint supports remain disjoint.

Our formalization. axiomatised
The authors' strategy chains the two preceding results: partition each atomless support into M prescribed-mass pieces, then use the ball-chain transport (Lemma B.1) to drive each piece into a small ball about its target atom x_k^i, handling the disjoint families one at a time so that — by the off-support identity (B.2) — the others stay fixed. Note the paper's proof uses only the **perceptron drift** ($\mathbf{V} \equiv 0$), so the statement honestly lives on the linear flow layer (`measureFlow`). It is a labeled `axiom`: an earlier proved assembly (partition + rotate + cluster per piece) was retired by the F14 restatement because it distributed a *mean-field* flow over the mixture — mixture linearity is meaningless for measure-dependent dynamics — and the faithful, paper-route proof through the now-discharged Lemma B.1 chain plus the atomless splitting (`exists_atomless_partition`, machine-checked) is the natural next discharge target.

```
/-- **Proposition 2.2** (clustering to a discrete measure). An atomless sphere-supported
probability measure can be driven `W₂`-close to a prescribed `M`-atom discrete measure.
AXIOM (`math.axiomatised`): the paper's V ≡ 0 gated route through Lemma B.1 is the
discharge target now that the chain is machine-checked. -/
axiom prop_2_2 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (hd : 3 ≤ d)
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (hatomless : ∀ x : Eucl d, μ {x} = 0)
    (hμsupp : supportedIn μ (sphere d))
    (M : ℕ) (x : Fin M → Eucl d) (hx : ∀ k, x k ∈ sphere d)
    (α : Fin M → ℝ≥0∞) (hα : ∑ k, α k = 1)
    (hα0 : ∀ k, α k ≠ 0)
    (ν_target : Measure (Eucl d))
    (htgt : ν_target = ∑ k : Fin M, α k • Measure.dirac (x k)) :
    ∃ θ : Params d, Axioms.W2 (measureFlow θ T μ) ν_target ≤ ε
```

Statements/MidLevel.lean

4 Disentangling supports

A piecewise-constant control separates the N overlapping input supports into pairwise-disjoint regions of the sphere, by steering each measure's *barycenter* into its own direction and then shrinking its geodesic hull around it. This disentanglement is the heart of the construction behind Theorems 1.1 and 1.2: once the supports are disjoint, each measure can be clustered and matched independently.

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §3, p. 14).
"We show that flows generated by self-attention can disentangle measures with overlapping supports—it actually suffices to consider $\mathbf{B} \equiv 0$." Set $\mathbb{Q}_1^{d-1} := \mathbb{S}^{d-1} \cap (\mathbb{R}_{>0})^d$ and

$$\mathsf{v}[\mu](t,x) = \mathbf{P}_x^{\perp}(\mathbf{V}(t)\, \mathbb{E}_{\mu(t)}[z] + \mathbf{W}(t)(\mathbf{U}(t)x + b(t))_+). \tag{3.1}$$

The whole section uses this reduced field (3.1): a perceptron term $\mathbf{W}(\mathbf{U}x+b)_+$ and a barycentric attention term $\mathbf{V}\,\mathbb{E}_\mu[z]$, both projected onto the tangent space.

4.1 Transport to the orthant

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.2, p. 15).
Suppose $T > 0$ and $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$ with $\bigcup_i \operatorname{supp} \mu_0^i \subsetneq \mathbb{S}^{d-1}$. There exists a piecewise constant $\mathbf{W} : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})$, having at most one switch and satisfying

$$\|\mathbf{W}\|_{L^\infty((0,T);\mathcal{M}_{d\times d}(\mathbb{R}))} \leq C/T$$

for some $C = C(N) > 0$, such that for any $i \in [\![1, N]\!]$ the solution μ^i to (1.3)–(1.2) with data μ_0^i and $\mathbf{V} \equiv \mathbf{B} \equiv \mathbf{U} \equiv 0$, $b \equiv \mathbf{1}$, satisfies $\operatorname{supp} \mu^i(T) \subset \mathbb{Q}_1^{d-1}$.
Proof. *(proof sketch; full proof p. 15.)* Pick a direction $\omega \in \mathbb{S}^{d-1} \setminus \bigcup_i \operatorname{supp} \mu_0^i$ not filling the sphere and set $\mathbf{W}_1\mathbf{1} = -\omega$. The characteristics solve $\dot{x} = -\mathbf{P}_x^\perp\omega$, so $\frac{d}{dt}\langle x, \omega\rangle = -1+\langle x, \omega\rangle^2 < 0$ off $\{\pm\omega\}$, hence by some T_0 all supports lie in $B(-\omega, \pi/8)$. Pick $\alpha \in \mathbb{Q}_1^{d-1}$ with $d_g(-\omega, -\alpha) > \pi/8$ (so $-\alpha$ is off every support) and on $[T_0, T]$ take $\mathbf{W}_2\mathbf{1} = \alpha$: now $\frac{d}{dt}\langle x, \alpha\rangle = 1 - \langle x, \alpha\rangle^2 \geq 0$ drives every point into a prescribed small cap around $\alpha \in \mathbb{Q}_1^{d-1}$. One switch; the norm bound follows by time-rescaling the two phases. $\square$

Our formalization. axiomatised
The authors' construction (attributed to them) is the two-phase control above: a first phase $\mathbf{W}_1\mathbf{1} = -\omega$ contracts every support toward $-\omega$ through the gate-free dynamics $\frac{d}{dt}\langle x, \omega\rangle = -(1 - \langle x, \omega\rangle^2)$ — the projector identity of leaf **L1** — and a second phase $\mathbf{W}_2\mathbf{1} = \alpha \in \mathbb{Q}_1^{d-1}$ drives them into a small cap inside the orthant. The dynamics here is pure perceptron ($\mathbf{V} \equiv 0$), so the statement lives faithfully on the linear flow layer, as a labeled `axiom` carrying the paper's hypotheses in full: a **probability** measure, **sphere-supported**, with a **missing cap** (`MissingCap`, the positive-gap encoding of $\bigcup \operatorname{supp} \subsetneq \mathbb{S}^{d-1}$; without it a dense-support measure refutes the statement — a homeomorphism preserves density). The budget is `switches θ ≤ 2` in the formal pieces convention (the paper's "one switch" counts the single discontinuity between the two constant phases).

```
/-- The support misses a spherical cap: some unit `ω` has a positive gap `δ` with
`⟪ω, x⟫ ≤ 1 - δ` on the full mass of `μ` (the faithful encoding of `supp μ ⊊ S^{d-1}`).
 ↪  -/
def MissingCap (μ : Measure (Eucl d)) : Prop :=
  ∃ ω : Eucl d, ‖ω‖ = 1 ∧ ∃ δ : ℝ, 0 < δ ∧ supportedIn μ {x | ⟪ω, x⟫ ≤ 1 - δ}

/-- **Lemma 3.2** (transport into the orthant). Two constant phases move a
 ↪  sphere-supported
probability measure with a missing cap into `Q₁^{d-1}`. AXIOM (`math.axiomatised`). -/
axiom lemma_3_2 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (T : ℝ) (hT : 0 < T)
    (hμs : supportedIn μ (sphere d)) (hgap : MissingCap μ) :
    ∃ θ : Params d, switches θ ≤ 2 ∧ supportedIn (measureFlow θ T μ) (orthant d)
```

Statements/MidLevel.lean

4.2 Shrinking a hull to its barycenter direction

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.3, p. 16; proof App. B.2, p. 33).
Let $\mu_0^i \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ be such that $\mathbb{E}_{\mu_0^i}[x]$ is not colinear with $\mathbb{E}_{\mu_0^j}[x]$ for $i \neq j$. Then for any $T > 0$, $\varepsilon > 0$, and $\nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ such that $\mathbb{E}_{\nu_0}[x]$ is colinear with $\mathbb{E}_{\mu_0^i}[x]$, there exists a piecewise constant $\theta : [0, T] \to \Theta$ having at most $O(d \cdot N)$ switches such that

$$\operatorname{conv}_g \operatorname{supp} \nu(T) \cup \operatorname{conv}_g \operatorname{supp} \mu^N(T) \subset B\Big(\mathbb{E}_{\mu_0^j}[z] \Big/ \|\mathbb{E}_{\mu_0^j}[z]\|,\ \varepsilon\Big)$$

and $\mu^i(T) = \mu_0^i$ for $i \neq j$, where μ^i, ν denote the unique solutions to (1.3)–(3.1) corresponding to data μ_0^i, ν_0, and parameters θ.
Proof. *(proof sketch; full proof App. B.2, p. 33.)* Three steps. **(1) Isolating** μ_0^N, ν_0: with $\mathbf{W} \equiv 0$ and $\mathbf{V}(t) = \sum_k \alpha_k \alpha_k^\top \mathbf{1}_{[T_{k-1}, T_k]}$, where $\{\alpha_k\}$ is an orthonormal basis of $\operatorname{span}(\mathbb{E}_{\mu_0^N}[z])^\perp$, the barycenter ODE (B.9) $\frac{d}{dt}\langle x, \alpha_1\rangle = \langle \mathbb{E}_{\mu(t)}[x], \alpha_1\rangle(1 - \langle \alpha_1, x\rangle^2)$ sends $x(t) \to \pm\alpha_k$, concentrating $\operatorname{supp} \mu(T_{d-1})$ near $\bigcup_k B(\pm\alpha_k, C_k\varepsilon_k) \subset \mathbb{S}^{d-1} \setminus \mathbb{Q}_1^{d-1}$ (flow Ψ_1, fixing μ_0^j for $j \leq N-1$). **(2) Clustering:** a separating half-space $\langle a, x\rangle + \underline{b}$ with $\mathbf{U} = \mathbf{1}a^\top$, $\mathbf{W}_2\mathbf{1} = \mathbb{E}_{\mu_0^N}[x]$ drives both supports into $B(\mathbb{E}_{\mu_0^N}[x]/\|\mathbb{E}_{\mu_0^N}[x]\|, \delta)$ (flow Ψ_2). **(3) Flow reversal:** $\Phi_{\text{fin}} := \Psi_1^{-1} \circ \Psi_2 \circ \Psi_1$ returns μ_0^i to itself for $i \leq N-1$ and clusters ν, μ^N within $B(\,\cdot\,, C_T\delta)$; pick δ small. $\square$

Our formalization. axiomatised
The authors' three-step argument (App. B.2) hinges on the **barycenter ODE (B.9)**, which is exactly our machine-checked leaf **L6** (`barycenter_hasDerivAt_inner`): along the attention field, $\frac{d}{dt}\langle x, \alpha\rangle = c\,(1 - \langle x, \alpha\rangle^2)$ with strict increase for $c > 0$, which is what drives each point to $\pm\alpha$ and lets the support be concentrated into an ε-ball around the barycenter direction. The drift is barycentric attention ($\mathbf{V} \neq 0$), i.e. genuinely **measure-dependent**, so the statement lives on the mean-field attention layer (`AttnSchedule`, `attnMeasureFlow`; the F14 restatement) as a labeled `axiom`, carrying the paper's context in full: a probability measure supported on the sphere inside the orthant (where Lemma 3.2 has placed it), concentrated to a ball around a unit direction.

```
/-- **Lemma 3.3** (shrink a measure's hull toward its barycenter direction). A
 ↪  sphere-supported
probability measure inside the orthant can be concentrated into an `ε`-ball around a
 ↪  unit
direction by the barycentric-attention dynamics. AXIOM (`math.axiomatised`). -/
axiom lemma_3_3 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (T ε : ℝ) (hT : 0 < T)
 ↪  (hε : 0 < ε)
    (hμs : supportedIn μ (sphere d)) (hμo : supportedIn μ (orthant d)) :
    ∃ (θ : AttnSchedule d) (α : Eucl d), AttnSchedule.durationSum θ = T ∧ α ∈ sphere d ∧
      supportedIn (attnMeasureFlow θ μ) (Metric.ball α ε)
```

Statements/MidLevel.lean

4.3 Making barycenters non-colinear

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 3.4, p. 16; proof App. B.3, p. 35).
Let $T > 0$ and let $\mu_0, \nu_0 \in \mathcal{P}(\mathbb{Q}_1^{d-1})$ be two different measures such that $\mathbb{E}_{\mu_0}[x] = \gamma_1 \mathbb{E}_{\nu_0}[x]$ for some $\gamma_1 \in (0, 1]$.

1. If $\gamma_1 = 1$, then, setting $\mathbf{V} \equiv 0$, there exist $\mathbf{W}, \mathbf{U} \in \mathcal{M}_{d\times d}(\mathbb{R})$ and $b \in \mathbb{R}^d$ such that the solutions μ, ν to (1.3)–(3.1) corresponding to μ_0, ν_0 and these parameters, satisfy $\mathbb{E}_{\mu(T)}[x] \neq \mathbb{E}_{\nu(T)}[x]$. Moreover the Lipschitz-continuous and invertible flow map $\Phi^T : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ induced by the characteristics satisfies $\Phi^T(x) = x$ for $x \in \mathbb{S}^{d-1} \setminus (\operatorname{conv}_g \operatorname{supp} \mu_0 \cup \operatorname{conv}_g \operatorname{supp} \nu_0)$. (3.2)
2. If $\gamma_1 \neq 1$, then, setting $\mathbf{B} \equiv 0$, there exist piecewise-constant $(\mathbf{V}, \mathbf{W}, \mathbf{U})$ and b with at most 2 switches such that $\mathbb{E}_{\mu(T)}[x] \neq \gamma_2\, \mathbb{E}_{\nu(T)}[x]$ for all $\gamma_2 \in \mathbb{R}$ (non-colinear).

Proof (Part 1). *(proof sketch; full proof App. B.3, p. 35.)* Since $\mu_0 \neq \nu_0$, there is an open ball $\mathcal{B} \subset \operatorname{supp} \mu_0 \cup \operatorname{supp} \nu_0$ with $\mu_0(\mathcal{B}) \neq \nu_0(\mathcal{B})$. There exists $x^* \in \mathcal{B}$ with

$$\mu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\mu_0(\mathrm{d}x) \;\neq\; \nu_0(\mathcal{B})x^* + \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x\,\nu_0(\mathrm{d}x),$$

for otherwise $x^* = \frac{1}{\mu_0(\mathcal{B})-\nu_0(\mathcal{B})} \int_{\mathbb{S}^{d-1}\setminus\mathcal{B}} x(\nu_0 - \mu_0)(\mathrm{d}x)$ would be forced constant over all of $\mathcal{B}$ — impossible. Taking such x^*, a perceptron block $\mathbf{U} = -\mathbf{1}a^\top$, $b = R\mathbf{1}$ (a, R the center and radius of $\mathcal{B}$), and $\mathbf{W}\mathbf{1} = x^*$, by Lemma B.2 pushes $\mu(T)$ to $\mu_0(\mathcal{B})\delta_{x^*} + \mu_0(\cdot \setminus \mathcal{B})$; continuity of the expectation in W_2 yields $\mathbb{E}_{\Phi_\# \mu_0}[x] \neq \mathbb{E}_{\Phi_\# \nu_0}[x]$, with $\Phi(x) = x$ off $\mathcal{B}$. $\square$

Our formalization. open
In the $\gamma_1 = 1$ (perceptron) case the authors locate a ball $\mathcal{B}$ on which μ_0, ν_0 differ in mass, then argue that **some $x^* \in \mathcal{B}$ makes the barycenters differ** — for if every x^* gave equality, x^* would be pinned to one value on the whole ball, which no map can be. That "no map is constant on a nonempty open ball" core is our machine-checked leaf **L10** (`exists_ne_in_ball`). We state the $\gamma_1 = 1$ conclusion as a labeled `axiom` on the linear layer (the paper sets $\mathbf{V} \equiv 0$ here), **carrying the paper's hypotheses in full**: μ, ν are distinct probability measures, supported on the sphere inside the orthant, with equal barycenters. Every hypothesis is load-bearing: the proof opens with "since $\mu_0 \neq \nu_0$" ($\mu = \nu$ refutes the bare statement — finding F11), and without sphere support a heavy-tailed orthant measure has the junk Bochner barycenter 0, satisfying the hypotheses vacuously (finding F12). (The attention case $\gamma_1 \neq 1$, at most two switches, is the companion axiom `lemma_3_4_part2`, on the same layer with a genuine $\gamma \in (0, 1)$ colinearity hypothesis.)

```
/-- **Lemma 3.4, Part 1** (`γ₁ = 1` case). Two distinct sphere-supported probability
 ↪ measures on
the orthant with equal barycenters can be driven, by a constant parameter, to differing
barycenters. AXIOM (`math.axiomatised`); the pigeonhole core is leaf L10. -/
axiom lemma_3_4_part1 (μ ν : Measure (Eucl d)) [IsProbabilityMeasure μ]
 ↪ [IsProbabilityMeasure ν]
    (T : ℝ) (hT : 0 < T) (hne : μ ≠ ν)
    (hμs : supportedIn μ (sphere d)) (hνs : supportedIn ν (sphere d))
    (hμ : supportedIn μ (orthant d)) (hν : supportedIn ν (orthant d))
    (hbar : barycenter μ = barycenter ν) :
    ∃ θ : Params d, barycenter (measureFlow θ T μ) ≠ barycenter (measureFlow θ T ν)
```

Statements/MidLevel.lean

4.4 Disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 3.1, pp. 15-17). Let $T > 0$ and $\mu_0^i \in \mathcal{P}(\mathbb{Q}_1^{d-1})$. There exists a piecewise constant $\theta : [0, T] \to \Theta$, having at most $O(d \cdot N)$ switches, such that for all $i \in [\![1, N]\!]$, the solution μ^i to (1.3)–(3.1) with data μ_0^i and θ satisfies

$$\operatorname{conv}_g \operatorname{supp} \mu^i(T) \cap \operatorname{conv}_g \operatorname{supp} \mu^j(T) = \emptyset \quad \text{if } i \neq j.$$

Proof. *(proof sketch; full proof pp. 16-17.)* Induction on N; the base case $N = 1$ is trivial. Assume the first $N - 1$ measures already have pairwise-disjoint geodesic hulls (3.3) and let μ_0^N be arbitrary. Since $\operatorname{supp} \mu_0^i \subset \mathbb{Q}_1^{d-1}$, (3.3) implies $\mathbb{E}_{\mu_0^i}[x]$ is not colinear with $\mathbb{E}_{\mu_0^j}[x]$ for $i \neq j \leq N - 1$. If $\mathbb{E}_{\mu_0^N}[x]$ is non-colinear with all of them, one application of Lemma 3.3 (small ε) shrinks μ^N's hull until it separates. Otherwise $\mathbb{E}_{\mu_0^N}[x]$ is colinear with $\mathbb{E}_{\mu_0^i}[x]$ for at most one i (WLOG $i = N - 1$), and four phases of $[0, T]$ finish it: (1) $[0, T/4]$ Lemma 3.3 peels μ^N, μ^{N-1} off the rest; (2) $[T/4, T/2]$ Lemma 3.4 Part 1 makes $\mathbb{E}_{\mu^{N-1}}[x] \neq \mathbb{E}_{\mu^N}[x]$ while (3.2) keeps the others fixed; (3) $[T/2, 3T/4]$ Lemma 3.4 Part 2 makes the two barycenters non-colinear; (4) $[3T/4, T]$ Lemma 3.3, now applicable to all N measures, separates them. Total $O(d \cdot N)$ switches. □

Our formalization. axiomatised
Disentanglement is where the measure dependence of self-attention is **essential**: the paper's own eq. (1.7) proves a measure-independent (linear) flow cannot separate overlapping inputs — two equal measures are carried to the same image by any single map. The fidelity audit (finding F14) showed the earlier linear statement of this node was refutable for exactly that reason, so the dynamical content now lives on the mean-field attention layer as the axiom `exists_disentangling_balls`: under a **shared missing direction** (the positive-gap form of (1.4)) and **pairwise distinct** inputs (the paper's standing assumption), one attention schedule concentrates each measure into its own ball around a unit direction, `2r`-separated, together with the per-member transport-map form of (B.2). `prop_3_1` is then **proved** from it: the carrier balls are pairwise disjoint (`Metric.ball_disjoint_ball`) and each lies in an open hemisphere (Cauchy-Schwarz) — the geometric packaging the paper leaves implicit.

Its badge is `axiomatised` because the effective status inherits the axiom: `#print axioms prop_3_1` lists exactly `exists_disentangling_balls` and the mean-field well-posedness axiom `exists_meanFieldFlow`.

```
/-- Shared missing direction (eq. 1.4, positive-gap form): one unit `ω` and one gap `δ >
 ↪  0`
work for every member of the family. -/
def SharedMissingDirection {N : ℕ} (μ : Fin N → Measure (Eucl d)) : Prop :=
  ∃ ω : Eucl d, ‖ω‖ = 1 ∧ ∃ δ : ℝ, 0 < δ ∧ ∀ i, supportedIn (μ i) {x | ⟪ω, x⟫ ≤ 1 - δ}

/-- **Proposition 3.1** (disentanglement). Under a shared missing direction and
 ↪  pairwise distinct
inputs, one attention schedule renders the supports pairwise disjoint, each in an open
hemisphere. PROVED over `exists_disentangling_balls` (effective `math.axiomatised`). -/
theorem prop_3_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (T : ℝ) (hT : 0 < T)
    (hμ : ∀ i, IsProbabilityMeasure (μ₀ i))
    (hμs : ∀ i, supportedIn (μ₀ i) (sphere d))
    (hne : Pairwise fun i j ⇒ μ₀ i ≠ μ₀ j)
    (hmiss : SharedMissingDirection μ₀) :
    ∃ θ : AttnSchedule d, AttnSchedule.durationSum θ = T ∧
      DisjointSupports (fun i ⇒ attnMeasureFlow θ (μ₀ i)) ∧
     ∀ i, ∃ e : Eucl d, ‖e‖ = 1 ∧ supportedIn (attnMeasureFlow θ (μ₀ i)) {x | 0 < ⟪e, x⟫}
```

Statements/MainResults.lean

What we strengthen. *(finding F2.)* The induction above opens by asserting that **disjoint geodesic hulls imply non-colinear barycenters** —

$$\operatorname{conv}_g \operatorname{supp} \mu_0^i \cap \operatorname{conv}_g \operatorname{supp} \mu_0^j = \emptyset \implies \mathbb{E}_{\mu_0^i}[x] \text{ not colinear with } \mathbb{E}_{\mu_0^j}[x]$$

— **without proof**. Our adversarial review flagged this as the one genuine rigor gap (finding F2). It is now the machine-checked leaf **L11** (`barycenter_noncolinear_of_disjoint_hull`, see the Self-contained leaves section) for the empirical regime of Theorem 1.1 / restricted Theorem 1.2, and **L11**′ (`barycenter_noncolinear_of_disjoint_hull_general`) for arbitrary probability measures via Mathlib's `Convex.integral_mem`. The barycenter, a nonnegative average of support points, lies in cone(supp); colinearity would put a common normalized direction in *both* hulls, contradicting disjointness. The headline `prop_3_1` stays `[axiomatised]{.badge .axiom}` only because it rests on the disentangling-dynamics axiom — not on this step, which is machine-checked. (Validated numerically by experiment E3.)
Leaves/BarycenterNonColinear.lean

5 Matching discrete measures

Once the disentanglement of §3 has rendered the supports pairwise disjoint and the clustering of §2 has collapsed each measure to a single point, matching reduces to steering finitely many points to their prescribed targets. A single perceptron gate — identically zero on a chosen parking cap and positive outside it — lets the flow drive one *active* point along a geodesic corridor to its target while every other point, parked inside the cap, stays fixed.

5.1 Proposition 4.2 — steer the active point

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.2, p. 18). Suppose $d \geqslant 3$. Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \qquad \text{for } i \in [\![1, M]\!], \tag{$\mathcal{D}$}$$

with $x_0^i \neq x_0^j$ and $y^i \neq y^j$ for $i \neq j$, with $x_0^i = y^i$ for $i \in [\![1, M-1]\!]$, and suppose that there exist $\gamma \in \mathbb{S}^{d-1}$ and $\varepsilon > 0$ such that $\langle \gamma, x_0^M - y^M \rangle = 0$ and $x_0^i \notin H_\varepsilon^\gamma$ for all $i \in [\![1, M-1]\!]$, where $H_\varepsilon^\gamma := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma \rangle| \leqslant \varepsilon\}$.
For any $T > 0$ there exist piecewise constant $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathscr{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, having at most 6 switches, such that for any i, the solution $x^i(\cdot)$ to

$$\begin{cases} \dot{x}^i(t) = \mathbf{P}_x^\perp \mathbf{W}(t)\big(\mathbf{U}(t)x^i(t) + b(t)\big)_+ & \text{in } [0, T] \\ x^i(0) = x_0^i, \end{cases} \tag{4.1}$$

satisfies $x^i(T) = y^i$. Moreover, there exists $C > 0$, not depending on $\mathcal{D}$ and T, such that

$$\|\theta\|_{L^\infty((0,T);\Theta)} \leqslant \frac{C}{T \cdot \varepsilon}.$$

Proof sketch (proof sketch; full proof pp. 18-23). The control is a 6-interval step function $(\mathbf{W}(t), \mathbf{U}(t), b(t)) = \sum_{j=1}^6 (\mathbf{W}_j, \mathbf{U}_j, b_j)\, \mathbf{1}_{[(j-1)T/6,\, jT/6]}(t)$ with $\mathbf{U}_5 = \mathbf{U}_1$, $\mathbf{U}_6 = \mathbf{U}_2$, $\mathbf{U}_3 = \mathbf{U}_4$, $b_5 = b_1$, $b_6 = b_2$, $b_3 = b_4$, $\mathbf{W}_5 = -\mathbf{W}_1$, $\mathbf{W}_6 = -\mathbf{W}_2$, i.e. the last two intervals time-reverse the first two — at most 6 switches. *Anchors (Step 1).* Since $\langle \gamma, x_0^M - y^M \rangle = 0$ one finds $\omega \in \mathbb{S}^{d-1}$ with $\langle \gamma, \omega \rangle = 0$ and $d_g(\omega, x_0^M), d_g(\omega, y^M) \geqslant \pi/2$; the points $\omega_\pm$ at geodesic distance $\pi/8$ from ω toward $\pm\gamma$ then satisfy $d_g(\omega_\pm, x_0^M), d_g(\omega_\pm, y^M) \geqslant 3\pi/8$, so $\{\langle \omega, x \rangle = \cos(\pi/8 + \tau)\}$ separates the cap $B(\omega, \pi/8 + \tau)$ from x_0^M, y^M. *Phase 1 — "gather" ψ_1 (Steps 1-2, 2 switches).* Two opposite gates $(\langle \pm\gamma, x \rangle - \epsilon/2)_+$ with drift targets $\omega_\pm$ run the Riemannian gradient flow of $d_g(\cdot, \omega_\pm)$ (using $\nabla_1 d_g(x, \omega_+) = -\mathbf{P}_x^\perp \omega_+ / \sqrt{1 - \langle x, \omega_+ \rangle^2}$); $\langle x, \omega_\pm \rangle$ increases monotonically (4.5) and the trajectory settles exponentially near $\omega_\pm$ by LaSalle / Hartman-Grobman (4.6), so after time $T/3$ every inactive point lies in the cap $B(\omega, 3\pi/16)$ while points of H_ε^γ stay fixed (4.7). *Phase 2 — "corridor" ψ_2 (Step 3, 2 switches).* The gate $(\langle -\omega, x \rangle + \cos(3\pi/16))_+$ is identically 0 on $B(\omega, 3\pi/16)$ and positive outside; driving x_0^M first toward an intermediate z_1 then toward z_2 along a geodesic corridor that lies outside the cap and passes through y^M lands the active point at y^M, while the inactive points do not move (gate off on the cap). *Phase 3 — "restore" ψ_1^{-1} (Step 4, 2 switches).* Re-running the gather action with flipped drifts time-reverses Phase 1, returning every inactive point to its start; the active point remains at y^M since the corridor gate is off on the cap. The composition $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$ sends $x_0^M \mapsto y^M$ and fixes all other x_0^i; choosing drift size $\|\mathbf{W}\| \asymp (T\varepsilon)^{-1}$ gives total time T and $\|\theta\|_{L^\infty} \lesssim 1/(T\varepsilon)$.

Our formalization. open
The authors' construction (pp. 18-23) realises the matching of one point as the gather / corridor / restore composition $(\psi_1)^{-1} \circ \psi_2 \circ \psi_1$, a 6-interval piecewise-constant control whose

middle phase steers the active x_0^M to y^M while a gate parked on the cap $B(\omega, 3\pi/16)$ freezes the inactive points. We state this as a labeled `axiom` over the linear flow layer (the paper's own dynamics here has $\mathbf{V} \equiv 0$, so the layer is faithful): with $d \geqslant 3$, points and targets **on the sphere**, **injective** inputs and targets (both restored fidelity hypotheses: the flow map is a bijection, so colliding targets are unreachable, and off-sphere points contradict the kernel-checked sphere invariance), and the inactive points already at their targets (`hfix`), at most 6 switches drive every input to its target while keeping the inactive ones fixed. Two of the construction's cores are discharged as kernel-checked leaves: the separating-hyperplane estimate of Step 1 is leaf **L3** (`Leaves/SeparatingHyperplane.lean`, `separating_hyperplane`), and the geodesic-gradient identity driving the gather and corridor flows is leaf **L4** (`Leaves/GeodesicGradient.lean`, `geodesicDist_hasDerivAt`). The selective-motion mechanism — active point reaches the target, parked points stay fixed — is validated numerically by experiment **E4**.

```
/-- **Proposition 4.2** (steer one active point). With `d ≥ 3`, on-sphere injective
↪  inputs and
targets, and the inactive points already at their targets, at most `6` switches move
↪  every input
to its target, keeping the inactive ones fixed. AXIOM (`math.axiomatised`): the
gather/corridor/restore construction is a geodesic gradient flow (leaves L3, L4). -/
axiom prop_4_2 (hd : 3 ≤ d) (M : ℕ) (x₀ y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)
    (hx₀s : ∀ i, x₀ i ∈ sphere d) (hys : ∀ i, y i ∈ sphere d)
    (hx₀ : Function.Injective x₀) (hy : Function.Injective y)
    (hfix : ∀ i : Fin M, (i : ℕ) < M - 1 → x₀ i = y i) :
    ∃ θ : Params d, switches θ ≤ 6 ∧ ∀ i, flowMap θ T (x₀ i) = y i
```

`Statements/MidLevel.lean`

5.2 Proposition 4.1 — match the ensemble

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, §4, Proposition 4.1, p. 18). Suppose $d \geqslant 3$. Consider

$$(x_0^i, y^i) \in \mathbb{S}^{d-1} \times \mathbb{S}^{d-1} \qquad \text{for } i \in [\![1, M]\!], \tag{$\mathcal{D}$}$$

with $x_0^i \neq x_0^j$ and $y^i \neq y^j$ for $i \neq j$, and suppose that for any i there exist $\gamma_i \in \mathbb{S}^{d-1}$ and $\varepsilon_i > 0$ such that $\langle \gamma_i, x_0^i - y^i \rangle = 0$ and $x_0^j \notin H_{\varepsilon_i}^{\gamma_i}$ for $j \neq i$, where

$$H_{\varepsilon_i}^{\gamma_i} := \{x \in \mathbb{S}^{d-1} : |\langle x, \gamma_i \rangle| \leqslant \varepsilon_i\}.$$

For any $T > 0$ there exist piecewise constant $\theta = (\mathbf{W}, \mathbf{U}, b) : [0, T] \to \mathcal{M}_{d\times d}(\mathbb{R})^2 \times \mathbb{R}^d$, having at most $6M$ switches, such that for any i, the solution $x^i(\cdot) \in \mathcal{C}^0([0, T]; \mathbb{S}^{d-1})$ to (4.1) satisfies $x^i(T) = y^i$. Moreover, there exists $C > 0$, not depending on $\mathcal{D}$ nor T, such that

$$\|\theta\|_{L^\infty((0,T);\Theta)} \leqslant \frac{C \cdot M}{T \min_i \varepsilon_i}.$$

Proof sketch (proof sketch; full proof p. 18). "The proof of Proposition 4.1 follows directly from [Proposition 4.2], combined with a straightforward induction argument": activating one pair (x_0^i, y^i) at a time, each application of Proposition 4.2 steers the i-th point to its target in at most 6 switches while leaving the already-matched points fixed, so M applications cost at most $6M$ switches and the norm bound accumulates to $C \cdot M/(T \min_i \varepsilon_i)$.

Our formalization. axiomatised
The authors obtain Proposition 4.1 by induction over the M pairs, each step being one application of Proposition 4.2 (the active point reaches its target; the previously matched points are inactive and stay fixed), giving the $\leqslant 6M$-switch budget and the $O(CM/(T \min_i \varepsilon_i))$ norm. We **prove** exactly that induction in Lean: base case the identity schedule; each step composes one `prop_4_2` application via `comp`/`flowMap_comp`, the sphere membership of the moved points carried by the kernel-checked `flowMap_mem_sphere`, the budget by `switches_comp`. The badge is `axiomatised` because the effective status inherits `prop_4_2`; `#print axioms prop_4_1` lists exactly `prop_4_2` and nothing else. (Validated numerically by experiment **E4**.)

```
/-- **Proposition 4.1** (match an ensemble). With `d ≥ 3` and distinct on-sphere
 ↪  inputs/targets,
at most `6M` switches steer every `x₀ i` to `y i`. PROVED by induction over `prop_4_2`
(effective `math.axiomatised`: its only axiom is `prop_4_2`). -/
theorem prop_4_1 (hd : 3 ≤ d) (M : ℕ) (x₀ y : Fin M → Eucl d) (T : ℝ) (hT : 0 < T)
    (hx₀s : ∀ i, x₀ i ∈ sphere d) (hys : ∀ i, y i ∈ sphere d)
    (hx₀ : Function.Injective x₀) (hy : Function.Injective y) :
    ∃ θ : Params d, switches θ ≤ 6 * M ∧ ∀ i, flowMap θ T (x₀ i) = y i := by
  induction M with /- identity schedule; then one prop_4_2 step per pair -/
```

Statements/MidLevel.lean

6 Main results

The paper assembles its leaves and mid-level propositions into one solution map Φ^T_{fin} by a three-stage recipe: **disentangle** the N supports until they are pairwise disjoint (Proposition 3.1), **cluster** each measure — to a single point (Proposition 2.1) or to an M-atom empirical measure (Proposition 2.2) — and **match** each cluster to its prescribed target (Propositions 4.1/4.2). Theorem 1.1 is the Dirac-target case, steering each input to a point mass δ_{x^i} with $O(d\,N)$ switches and an explicit norm bound, while Theorem 1.2 is the general case, reaching any L^2 transport-matchable targets by interposing the map-transfer Lemma 5.1, the flow-approximation Lemma 5.4, and the coupling bound Lemma 5.2. Both theorems are **proved by assembly** in Lean over the labeled axiom layer, on the mean-field attention dynamics (`AttnSchedule`, `attnMeasureFlow`) that the F14 fidelity finding showed is the honest home for measure-dependent statements; their effective status is `axiomatised`, the minimum over their documented axiom surface. Proposition 6.2 — the generic $O(1)$-switch disentanglement heuristic — is reviewed on paper only.

6.1 Lemma 5.1 — transport maps preserved by disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.1, p. 24).
Suppose that for every i there exists $\mathsf{T}^i \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ with $\mathsf{T}^i_\# \mu_0^i = \mu_1^i$. Consider the flow map $\Phi_1 : \mathcal{P}(\mathbb{S}^{d-1}) \to \mathcal{P}(\mathbb{S}^{d-1})$ (resp. Φ_3) given by Proposition 3.1 with data $\mu_0^i \in \mathcal{P}(\mathbb{S}^{d-1})$ (resp. $\mu_1^i \in \mathcal{P}(\mathbb{S}^{d-1})$). Then there exists a Lipschitz-continuous and invertible map $\psi : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ such that

$$\psi\big|_{\mathrm{supp}\,\Phi_1(\mu_0^i)} = \psi^i\big|_{\mathrm{supp}\,\Phi_1(\mu_0^i)} \tag{5.1}$$

for any $i \in \{1, \ldots, N\}$, where $\psi^i : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ is another Lipschitz-continuous and invertible map that satisfies

$$(\psi^i \circ \mathsf{T}^i_{\Phi_1})_\# \mu_0^i = (\psi \circ \mathsf{T}^i_{\Phi_1})_\# \mu_0^i = (\mathsf{T}^i_{\Phi_3})_\# \mu_1^i = \Phi_3(\mu_1^i)$$

for some Lipschitz-continuous and invertible maps $\mathsf{T}^i_{\Phi_1}, \mathsf{T}^i_{\Phi_3} : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$. The proof can be found in Appendix B.4.

Our formalization. axiomatised
The paper's strategy (Geshkovski, Rigollet, Ruiz-Balet, App. B.4, p. 37) exploits disentanglement: once Proposition 3.1 has pushed the supports $\mathrm{supp}\,\Phi_1(\mu_0^i)$ pairwise disjoint, the per-pair transport maps T^i are carried through the flow and the individual matchings ψ^i are glued into one global ψ agreeing with each ψ^i on the i-th (now disjoint) support. It is a labeled `axiom`. The **disjoint-supports hypotheses on both families are load-bearing and kept explicit** (`DisjointSupports μ₀`, `DisjointSupports μ₁`): without them the gluing is genuinely impossible — with $\mu_0^0 = \mu_0^1 = \delta_a$ and distinct targets a single ψ would need two values at a (finding F11). The conclusion asserts a **measurable** ψ, not the paper's "invertible" one: invertibility is unsatisfiable even per pair (no injection pushes an atomless measure onto an atom), and the paper's own proof builds $\psi^i = \mathsf{T}^i_{\Phi_3} \circ \mathsf{T}^i \circ (\mathsf{T}^i_{\Phi_1})^{-1}$ from the non-invertible transport T^i — a statement/proof mismatch recorded as erratum candidate **E2** (finding F13).

```
/-- **Lemma 5.1** (transport map after disentanglement). If both families are
 ↪  disentangled (pairwise
disjoint supports, as Proposition 3.1 provides) and each pair is matchable, a single
 ↪  measurable map
matches them all. AXIOM (`math.axiomatised`); the paper's "invertible" is erratum
 ↪  candidate E2. -/
axiom lemma_5_1 {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
    (hdisj₀ : DisjointSupports μ₀) (hdisj₁ : DisjointSupports μ₁)
    (hmatch : ∀ i, ∃ Ti : Eucl d → Eucl d, (μ₀ i).map Ti = μ₁ i) :
    ∃ ψ : Eucl d → Eucl d, Measurable ψ ∧ ∀ i, (μ₀ i).map ψ = μ₁ i
```

Statements/MidLevel.lean

6.2 Lemma 5.4 — L² approximation by a flow map

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Lemma 5.4, p. 24).
Suppose $\varepsilon > 0$ and $\mu \in \mathcal{P}(\mathbb{S}^{d-1})$. For every $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$, there exists a Lipschitz-continuous and invertible map $\psi_\varepsilon : \mathbb{S}^{d-1} \to \mathbb{S}^{d-1}$ induced by the solution map of (B.1), namely $\Phi^T_{\theta_\varepsilon}(\mu) = (\psi_\varepsilon)_\# \mu$ for some piecewise constant $\theta_\varepsilon : [0,T] \to \Theta$ with finitely many switches, such that

$$\|\psi - \psi_\varepsilon\|_{L^2(\mu)} \leq \varepsilon.$$

The proof of Lemma 5.4 is involved, so we postpone it to Appendix B.5.

Our formalization. axiomatised
The paper (Geshkovski, Rigollet, Ruiz-Balet, App. B.5, p. 38) shows any $\psi \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ is approximated in $L^2(\mu)$, to any tolerance, by a flow map ψ_ε of the continuity equation with piecewise-constant control and finitely many switches; chained with the coupling bound $W_2(\mathsf{T}^1_\# \mu, \mathsf{T}^2_\# \mu) \leq \|\mathsf{T}^1 - \mathsf{T}^2\|_{L^2(\mu)}$ (Lemma 5.2, leaf L7) this converts L^2 control of the map into W_2 control of the pushforward. It is a labeled `axiom` on the mean-field layer, carrying the paper's hypotheses in full: a **probability** measure on the **sphere**, a **measurable, a.e. sphere-valued** ψ (the paper's $L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ — without sphere-valuedness the statement is refutable, since flow approximants are sphere-valued and cannot L^2-approach an off-sphere target; finding F12). The measurability and integrability of the displacement are made explicit so the `W₂` map bound can consume them.

```
/-- **Lemma 5.4** (`L²` approximation by a flow map). Any measurable, a.e. sphere-valued
 ↪ transport
map `ψ` is approximated in `L²(μ)` by a mean-field flow map, to any tolerance. AXIOM
(`math.axiomatised`). Combined with the coupling bound (leaf L7) this controls `W₂`. -/
axiom lemma_5_4 (μ : Measure (Eucl d)) [IsProbabilityMeasure μ] (ψ : Eucl d → Eucl d) (T
 ↪ ε : ℝ)
    (hT : 0 < T) (hε : 0 < ε)
    (hμs : supportedIn μ (sphere d)) (hψm : Measurable ψ)
    (hψs : ∀ᵐ x ∂μ, ψ x ∈ sphere d) :
    ∃ (θ : AttnSchedule d) (ψε : Eucl d → Eucl d),
      AttnSchedule.durationSum θ = T ∧
      attnMeasureFlow θ μ = μ.map ψε ∧ Measurable ψε ∧
      Integrable (fun x ↦ ‖ψ x - ψε x‖ ^ 2) μ ∧
      Real.sqrt (∫ x, ‖ψ x - ψε x‖ ^ 2 ∂μ) ≤ ε
```

Statements/MidLevel.lean

6.3 Theorem 1.2 — matching transport-matchable targets

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.2, p. 5).
Suppose $d \geq 3$. Consider data $(\mathcal{D})$ such that

(1) there exist $w_0, w_1 \in \mathbb{S}^{d-1}$ such that

$$w_0 \notin \bigcup_i \operatorname{supp}(\mu_0^i) \quad \text{and} \quad w_1 \notin \bigcup_i \operatorname{supp}(\mu_1^i); \tag{1.5}$$

(2) for any $i \in \{1, \dots, N\}$, there exists $\mathsf{T}^i \in L^2(\mathbb{S}^{d-1}; \mathbb{S}^{d-1})$ such that $\mathsf{T}^i_{\#}\mu_0^i = \mu_1^i$.

Then for any $T > 0$ and $\varepsilon > 0$, there exists $\theta \in L^\infty((0,T); \Theta)$ such that for any $i \in \{1, \dots, N\}$, the unique solution $\mu^i \in \mathcal{C}^0([0,T]; \mathcal{P}(\mathbb{S}^{d-1}))$ to (1.3) with data μ_0^i and parameters θ satisfies

$$\mathsf{W}_2(\mu^i(T), \mu_1^i) \leq \varepsilon.$$

Moreover, θ can be chosen piecewise constant.

Proof sketch (full proof §5.1, p. 25). The construction $\Phi^T_{\text{fin}} := (\Phi^{T/3}_{\theta_3})^{-1} \circ \Phi^{T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1}$ proceeds in three steps. **(1) Disentanglement:** Proposition 3.1 renders the input supports (μ_0^i) pairwise disjoint via Φ_{θ_1} on $[0, T/3]$ and the target supports (μ_1^i) via Φ_{θ_3} on $[2T/3, T]$; well-posedness and time-reversibility give a constant C with $\mathsf{W}_2((\Phi^T_{\theta_3})^{-1}\mu, (\Phi^T_{\theta_3})^{-1}\nu) \leq C\,\mathsf{W}_2(\mu, \nu)$ (5.3). **(2) Matching:** Lemma 5.1 produces a single Lipschitz-invertible ψ agreeing with each ψ^i on the disentangled support, and Lemma 5.4 realizes ψ up to L^2 error by a flow map $\psi_\varepsilon = \Phi_{\theta_2}$ on $[T/3, 2T/3]$; the coupling bound (Lemma 5.2) turns the L^2 error into $\mathsf{W}_2((\Phi^{2T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1})(\mu_0^i), \Phi^T_{\theta_3}(\mu_1^i)) \leq \varepsilon/C$ (5.5–5.6). **(3) Composition:** applying $(\Phi^T_{\theta_3})^{-1}$ and (5.3) yields $\mathsf{W}_2(\mu^i(T), \mu_1^i) \leq \varepsilon$ for all i.

Our formalization. axiomatised

The paper's proof (Geshkovski, Rigollet, Ruiz-Balet, §5.1) composes the three stages above: disentangle both families by Proposition 3.1, transfer the transport maps into a single ψ by Lemma 5.1, approximate ψ by a flow map by Lemma 5.4, and bound the resulting W_2 error by the coupling Lemma 5.2. We **prove** the theorem by assembly over the mean-field axiom layer: disentangle the inputs (`prop_3_1`), match each disentangled measure to its target through the per-member transport-map form of `exists_disentangling_balls`, approximate by `lemma_5_4`, bound with the machine-checked coupling lemma, and combine the per-member schedules with `exists_parked_schedule`. The `W₂` bookkeeping is machine-checked; `#print axioms theorem_1_2` lists exactly `exists_meanFieldFlow`, `exists_disentangling_balls`, `exists_parked_schedule`, `lemma_5_4`. The target missing direction (`_hmiss₁`, the second half of (1.5)) is retained in the statement for fidelity even though this assembly absorbs the target-disentanglement step into the parking axiom.

```
/-- **Theorem 1.2** (general targets). If every input/target pair is matchable by a
 ↪  measurable
sphere-valued transport map and the families share missing directions, a single
 ↪  attention schedule
steers each input to within `ε` of its target in `W₂`. PROVED by assembly (effective
`math.axiomatised`; footprint: exists_meanFieldFlow, exists_disentangling_balls,
exists_parked_schedule, lemma_5_4). -/
theorem theorem_1_2 (hd : 3 ≤ d) {N : ℕ} (μ₀ μ₁ : Fin N → Measure (Eucl d))
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε)
    (hmiss₀ : SharedMissingDirection μ₀) (_hmiss₁ : SharedMissingDirection μ₁)
    (hμ : ∀ i, IsProbabilityMeasure (μ₀ i))
    (hμ₀s : ∀ i, supportedIn (μ₀ i) (sphere d))
    (hne : Pairwise fun i j ⇒ μ₀ i ≠ μ₀ j)
    (hmatch : Matchable μ₀ μ₁) :
    ∃ θ : AttnSchedule d, AttnSchedule.durationSum θ = T ∧
      ∀ i, Axioms.W2 (attnMeasureFlow θ (μ₀ i)) (μ₁ i) ≤ ε
```

Statements/MainResults.lean

6.4 Theorem 1.1 — matching Dirac targets

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Theorem 1.1, p. 5). Suppose $d \geq 3$. Consider data $(\mathcal{D})$ such that

(1) there exists $w_0 \in \mathbb{S}^{d-1}$ such that

$$w_0 \notin \bigcup_i \operatorname{supp}(\mu_0^i); \tag{1.4}$$

(2) for any $i \in \{1, \dots, N\}$, we have $\mu_1^i = \delta_{x^i}$.

Then for any $T > 0$ and $\varepsilon > 0$, there exists $\theta \in L^\infty((0,T);\Theta)$ such that for any $i \in \{1, \dots, N\}$, the unique solution $\mu^i \in \mathcal{C}^0([0,T];\mathcal{P}(\mathbb{S}^{d-1}))$ to (1.3) with data μ_0^i and parameters θ satisfies

$$\mathsf{W}_2(\mu^i(T), \mu_1^i) \leq \varepsilon.$$

Moreover, θ can be chosen piecewise constant, with $O(d \cdot N)$ switches, and

$$\|\theta\|_{L^\infty((0,T);\Theta)} = O\left(\frac{d \cdot N}{T} + \log\frac{1}{\varepsilon}\right).$$

Proof sketch (full proof §5.2, p. 29). The proof follows the same ideas as Theorem 1.2 but is significantly simpler — some steps are omitted — using $\Phi^T_{\mathrm{fin}} := \Phi^T_{\theta_3} \circ \Phi^{2T/3}_{\theta_2} \circ \Phi^{T/3}_{\theta_1}$. **(1) Disentangle:** Φ_{θ_1} on $[0, T/3]$ from Proposition 3.1, with $\|\theta_1\| \lesssim d\,N/T$. **(2) Cluster:** Φ_{θ_2} on $[T/3, 2T/3]$ from Proposition 2.1, applicable since the $\Phi^{T/3}(\mu_0^i)$ are now disjoint and single-hemisphere, driving each to within δ of a point $\delta_{x_0^i}$ with $T_\delta = O(\log 1/\delta)$, so $\|\theta_2\| \lesssim \log 1/\delta$. **(3) Match:** Φ_{θ_3} on $[2T/3, T]$ from Proposition 4.1 steers the ensemble of atoms x_0^i to the targets y^i, with $\|\theta_3\| \lesssim N/T$; since all measures are δ-close to $\delta_{x_0^i}$ the error is $e^{O(N \cdot T)}\delta$. All in all $\mathsf{W}_2(\mu^i(T), \delta_{y^i}) \leq \varepsilon$ with $\|\theta\|_{L^\infty((0,T);\Theta)} = O(d \cdot N/T + \log 1/\varepsilon)$.

Our formalization. axiomatised

The paper's proof (Geshkovski, Rigollet, Ruiz-Balet, §5.2) is the simpler Dirac-target specialization: disentangle by Proposition 3.1 ($\|\theta_1\| \lesssim dN/T$), cluster each now-disjoint single-hemisphere measure to within δ of a point by Proposition 2.1 ($T_\delta = O(\log 1/\delta)$, $\|\theta_2\| \lesssim \log 1/\delta$), then match the ensemble of cluster atoms x_0^i to the targets y^i by Proposition 4.1 ($\|\theta_3\| \lesssim N/T$); Lemma 5.4 is not needed. The end-to-end construction is validated numerically by experiment **E6**, while the negative control **E7** shows that a single *linear* continuity equation cannot achieve it — the same obstruction that finding F14 turned into a kernel refutation of the earlier linear statement of this very node, and the reason the theorem now lives on the mean-field attention layer. We **prove** it by assembly: disentangle (`prop_3_1`), cluster each disentangled measure to its on-sphere target (`cluster_to_point`), and combine the per-member schedules (`exists_parked_schedule`); `#print axioms theorem_1_1` lists exactly `exists_meanFieldFlow`, `cluster_to_point`, `exists_disentangling_balls`, `exists_parked_schedule`. The paper's standing assumption of pairwise distinct inputs is the explicit hypothesis `hne` — with identical inputs no single flow can separate them. The $O(d\,N)$ switch count is deliberately deferred (the paper's constant is unspecified; inventing one risks a false axiom) and the norm bound is now *expressible* (attention blocks carry genuine operators) but not yet threaded — both documented in the roadmap, not silently encoded.

```
/-- **Theorem 1.1** (Dirac targets). If the targets are on-sphere point masses `δ_{x i}`
↪ and the
pairwise-distinct sphere-supported inputs share a missing direction, a single attention
↪ schedule
steers each input to within `ε` of its target in `W₂`. PROVED by assembly (effective
`math.axiomatised`; footprint: exists_meanFieldFlow, cluster_to_point,
exists_disentangling_balls, exists_parked_schedule). -/
theorem theorem_1_1 (hd : 3 ≤ d) {N : ℕ} (μ₀ : Fin N → Measure (Eucl d)) (x : Fin N →
↪ Eucl d)
    (T ε : ℝ) (hT : 0 < T) (hε : 0 < ε) (hmiss : SharedMissingDirection μ₀)
    (hμ : ∀ i, IsProbabilityMeasure (μ₀ i))
    (hμs : ∀ i, supportedIn (μ₀ i) (sphere d)) (hx : ∀ i, x i ∈ sphere d)
    (hne : Pairwise fun i j ⇒ μ₀ i ≠ μ₀ j) :
    ∃ θ : AttnSchedule d, AttnSchedule.durationSum θ = T ∧
      ∀ i, Axioms.W2 (attnMeasureFlow θ (μ₀ i)) (Measure.dirac (x i)) ≤ ε
```

Statements/MainResults.lean

6.5 Proposition 6.2 — generic O(1) disentanglement

From the original paper (Geshkovski, Rigollet, Ruiz-Balet, Proposition 6.2, p. 30). Let $d \geq 3$, $\pi = \mathrm{Unif}((\mathbb{S}^{d-1})^n)$, and sample $\mu_0^1, \ldots, \mu_0^N$ i.i.d. from

$$\mathcal{S}_n := \left\{ \frac{1}{n} \sum_{k=1}^{n} \delta_{x_k} : x_k \in \mathbb{S}^{d-1} \right\} \simeq (\mathbb{S}^{d-1})^n.$$

Consider (1.3)–(1.2) with $\mathbf{B} \equiv \beta I_d$ ($\beta \geq 0$), $\mathbf{V} \equiv I_d$, $\mathbf{W} \equiv 0$. Let x_*^i be the (a.s. existing) limit cluster point of $\mu^i(t)$. Then $\mathbb{P}[x_*^i \neq x_*^j] = 1$ for $i \neq j$.

Our formalization. informal
Reviewed on paper; not stated in Lean. The paper's argument (sketch of proof, p. 30) is a symmetry/measure-theoretic one: for π-almost every empirical input the dynamics with $\mathbf{B} \equiv \beta I_d$, $\mathbf{V} \equiv I_d$, $\mathbf{W} \equiv 0$ converge to a Dirac δ_X, defining a map $f : \mathcal{S}_n \to \mathbb{S}^{d-1}$. Because the flow and the law π are $\mathbb{O}(d)$-equivariant, the law $\nu = f_\# \pi$ of the limit is rotation-invariant, hence non-atomic — an atom at x_0 of mass p would force an atom of mass p at every $\mathsf{R}x_0$, contradicting $\nu(\mathbb{S}^{d-1}) = 1$ once enough distinct rotations are chosen. Two independent inputs then give independent limits with this common non-atomic law, so $\mathbb{P}[x_*^i = x_*^j] = \int \nu(\{x\})\, \mathrm{d}\nu(x) = 0$. We do not state this in Lean: it would require Haar measure on $\mathbb{O}(d)$, the equivariant ω-limit map f, and non-atomicity infrastructure outside our current axiom layer. The result is recorded as the heuristic that the generic switch complexity is $O(1)$ rather than the $O(d\,N)$ of Theorem 1.1.